

Major concepts for fitting logistic regression

- Regression coefficients, β 's, are non-linearly related to the mean!
 - Analytic (closed-form) solution does NOT exist!!!
 - Need an optimization algorithm to fit the model.
- Gradient can be efficiently calculated using the CHAIN RULE.
- Iterations of a Newton-based optimization method can be interpreted as a Weighted Least Squares problem.
 - Highlights the generalization of the linear modeling tools and concepts.
 - Coefficient posterior uncertainty still dictated by feature relationships!

The only unknowns in logistic regression are the unknown $\boldsymbol{\beta}$ -parameters

- The learning problem is therefore to update our belief about the β -parameters after collecting observations.

$$p(\boldsymbol{\beta} \mid \mathbf{y}, \mathbf{X}) \propto p(\mathbf{y} \mid \boldsymbol{\beta}, \mathbf{X})p(\boldsymbol{\beta})$$

We will use the notation that there are N observations of D features and the binary outcome which is encoded as $y_n = 1$ for the event and $y_n = 0$ for the non-event.

We saw that the log-posterior surface is unimodal and visually seemed relatively well approximated by a MVN distribution

- We will now go through deriving the MVN approximate posterior distribution on the unknown β -parameters.

With the Laplace Approximation, we are interested in the un-normalized log-posterior

$$\log[p(\boldsymbol{\beta} \mid \mathbf{y}, \mathbf{X})] \propto \log[p(\mathbf{y} \mid \boldsymbol{\beta}, \mathbf{X})] + \log[p(\boldsymbol{\beta})]$$

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- To simplify the derivations slightly, we will assume an infinitely diffuse prior, $\log[p(\boldsymbol{\beta})] \approx 0$, and so the log-posterior is approximately equal to the log-likelihood:

$$\log[p(\boldsymbol{\beta} \mid \mathbf{y}, \mathbf{X})] \approx \log[p(\mathbf{y} \mid \boldsymbol{\beta}, \mathbf{X})]$$

The following derivations also hold for Maximum Likelihood Estimation!

The log-likelihood is the sum of N conditionally independent Binomial (Bernoulli) likelihoods

- First, rewrite the log-likelihood in terms of the event probability, μ_n , rather than in terms of the inputs and coefficients.

$$\log[p(\mathbf{y} \mid \boldsymbol{\beta}, \mathbf{X})] \rightarrow \log[p(\mathbf{y} \mid \boldsymbol{\mu})] = \sum_{n=1}^N (\log[p(y_n \mid \mu_n)])$$

The log-likelihood is the sum of N conditionally independent Binomial (Bernoulli) likelihoods

- Substitute in the Bernoulli log-likelihood functional form.

$$\sum_{n=1}^N (\log[p(y_n | \mu_n)]) = \sum_{n=1}^N (y_n \log[\mu_n] + (1 - y_n) \log[1 - \mu_n])$$

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- The event probability, μ_n , is related to the linear predictor, η_n , through the inverse link function:

$$\mu_n = \text{logit}^{-1}(\eta_n)$$

Let's first consider the simple single input linearly related to the linear predictor

$$\eta_n = \beta_0 + \beta_1 x_n$$

Complete model formulation we are working with

$$\sum_{n=1}^N (\log[p(y_n | \mu_n)]) = \sum_{n=1}^N (y_n \log[\mu_n] + (1 - y_n) \log[1 - \mu_n])$$

$$\mu_n = \text{logit}^{-1}(\eta_n)$$

$$\eta_n = \beta_0 + \beta_1 x_n$$

The Laplace Approximation has two primary steps

- The first step is to find the posterior mode.
- The second step is to calculate the Hessian matrix at the posterior mode.

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- The first step is to find the posterior mode.

Because we assumed an infinitely diffuse prior, the posterior mode equals the Maximum Likelihood Estimate!!

- The second step is to calculate the Hessian matrix at the posterior mode.

To help with the book-keeping of the derivatives define the n -th observation's likelihood as:

$$L_n = y_n \log[\mu_n] + (1 - y_n) \log[1 - \mu_n]$$

- The “complete” log-likelihood is the sum over all N likelihoods:

$$L = \sum_{n=1}^N (L_n)$$

To find the posterior mode we must take the derivative of the log-likelihood wrt each parameter

$$\mathbf{g} = \begin{bmatrix} \frac{\partial L}{\partial \beta_0} \\ \frac{\partial L}{\partial \beta_1} \end{bmatrix} = \begin{bmatrix} \frac{\partial}{\partial \beta_0} \left(\sum_{n=1}^N (L_n) \right) \\ \frac{\partial}{\partial \beta_1} \left(\sum_{n=1}^N (L_n) \right) \end{bmatrix}$$

So we must calculate the n -th likelihood's contribution to the derivative

$$\mathbf{g} = \begin{bmatrix} \frac{\partial L}{\partial \beta_0} \\ \frac{\partial L}{\partial \beta_1} \end{bmatrix} = \begin{bmatrix} \frac{\partial}{\partial \beta_0} \left(\sum_{n=1}^N (L_n) \right) \\ \frac{\partial}{\partial \beta_1} \left(\sum_{n=1}^N (L_n) \right) \end{bmatrix} = \begin{bmatrix} \sum_{n=1}^N \left(\frac{\partial L_n}{\partial \beta_0} \right) \\ \sum_{n=1}^N \left(\frac{\partial L_n}{\partial \beta_1} \right) \end{bmatrix}$$

The “complete” derivative is the sum of the derivatives across all N observations

The partial derivatives will at first seem really complex...

- We have to calculate the partial derivatives of the n -th log-likelihood with respect to the intercept, β_0 , and the slope, β_1 .
- But why was the model written in terms of μ_n and η_n ?

$$\begin{aligned} L_n &= y_n \log[\mu_n] + (1 - y_n) \log[1 - \mu_n] \\ \mu_n &= \text{logit}^{-1}(\eta_n) \\ \eta_n &= \beta_0 + \beta_1 x_n \end{aligned}$$

So that way we can easily use the chain rule!

$$\begin{aligned}L_n &= y_n \log[\mu_n] + (1 - y_n) \log[1 - \mu_n] \\ \mu_n &= \text{logit}^{-1}(\eta_n) \\ \eta_n &= \beta_0 + \beta_1 x_n\end{aligned}$$

- Can write the partial derivatives in terms of three derivatives:

$$\frac{\partial L_n}{\partial \beta_0} = \frac{\partial L_n}{\partial \mu_n} \frac{\partial \mu_n}{\partial \eta_n} \frac{\partial \eta_n}{\partial \beta_0} \qquad \frac{\partial L_n}{\partial \beta_1} = \frac{\partial L_n}{\partial \mu_n} \frac{\partial \mu_n}{\partial \eta_n} \frac{\partial \eta_n}{\partial \beta_1}$$

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The first two derivatives are the same for both the intercept and slope

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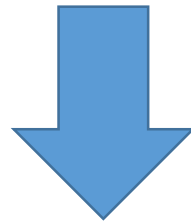
Start with the derivative of the n -th log-likelihood with respect to the probability of the event.

You have worked with this derivative already

$$\frac{\partial L_n}{\partial \mu_n} = \frac{\partial}{\partial \mu_n} (y_n \log[\mu_n] + (1 - y_n) \log[1 - \mu_n])$$

You have worked with this derivative already

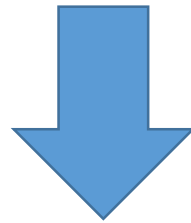
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$$\frac{\partial L_n}{\partial \mu_n} = y_n \frac{\partial}{\partial \mu_n} (\log[\mu_n]) + (1 - y_n) \frac{\partial}{\partial \mu_n} (\log[1 - \mu_n])$$

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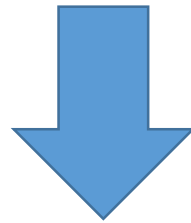
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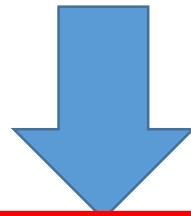
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$$\frac{\partial L_n}{\partial \mu_n} = \boxed{\frac{y_n}{\mu_n}} + (1 - y_n) \frac{\partial}{\partial \mu_n} (\log[1 - \mu_n])$$

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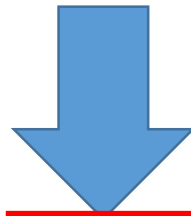
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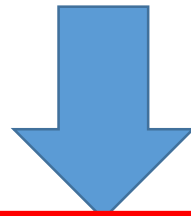
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$$\frac{\partial L_n}{\partial \mu_n} = \frac{y_n}{\mu_n} + (1 - y_n) \frac{(-1)}{1 - \mu_n}$$

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$$\frac{\partial L_n}{\partial \mu_n} = \frac{\partial}{\partial \mu_n} (y_n \log[\mu_n] + (1 - y_n) \log[1 - \mu_n])$$



$$\frac{\partial L_n}{\partial \mu_n} = \frac{y_n}{\mu_n} - \frac{1 - y_n}{1 - \mu_n}$$

So that way we can easily use the chain rule!

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Next consider the derivative of the event probability
with respect to the linear predictor

This derivative is more complicated...

- Requires taking the derivative through a non-linear function

$$\frac{\partial \mu_n}{\partial \eta_n} = \frac{\partial}{\partial \eta_n} (\mu_n) = \frac{\partial}{\partial \eta_n} (\text{logit}^{-1}(\eta_n)) = \frac{\partial}{\partial \eta_n} \left(\frac{\exp(\eta_n)}{1 + \exp(\eta_n)} \right)$$

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$$\frac{\partial \mu_n}{\partial \eta_n} = \mu_n(1 - \mu_n)$$

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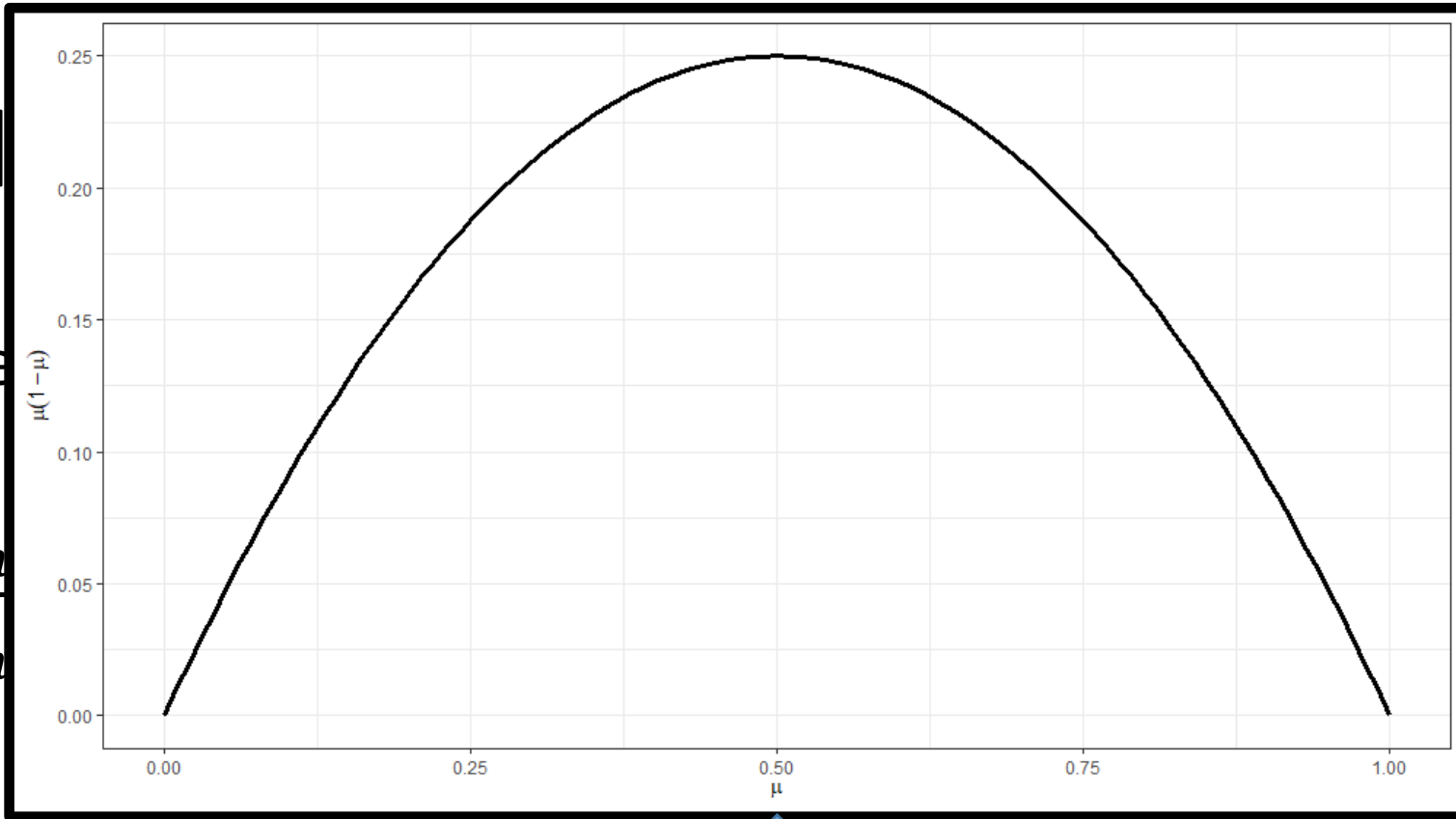
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Variance of a Bernoulli
random variable with
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This d

- Require

$$\frac{\partial \mu_n}{\partial \eta_n}$$



on

$$\frac{p(\eta_n)}{p(\eta_n)}$$

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Variance of a Bernoulli
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Next, consider the partial derivatives of the linear predictor with respect to the parameters

Partial derivatives of the linear predictor look like the derivatives of the linear model

$$\frac{\partial \eta_n}{\partial \beta_0} = \frac{\partial}{\partial \beta_0} (\beta_0 + \beta_1 x_n) = ?$$

$$\frac{\partial \eta_n}{\partial \beta_1} = \frac{\partial}{\partial \beta_1} (\beta_0 + \beta_1 x_n) = ?$$

Partial derivatives of the linear predictor look like the derivatives of the linear model

$$\frac{\partial \eta_n}{\partial \beta_0} = \frac{\partial}{\partial \beta_0} (\beta_0 + \beta_1 x_n) = 1$$

$$\frac{\partial \eta_n}{\partial \beta_1} = \frac{\partial}{\partial \beta_1} (\beta_0 + \beta_1 x_n) = x_n$$

Combine the terms

$$\frac{\partial L_n}{\partial \beta_0} = \underbrace{\frac{\partial L_n}{\partial \mu_n}}_{\text{blue}} \underbrace{\frac{\partial \mu_n}{\partial \eta_n}}_{\text{yellow}} \underbrace{\frac{\partial \eta_n}{\partial \beta_0}}_{\text{orange}} = \underbrace{\left(\frac{y_n}{\mu_n} - \frac{1 - y_n}{1 - \mu_n} \right)}_{\text{blue}} \times \underbrace{(\mu_n(1 - \mu_n))}_{\text{yellow}} \times \underbrace{1}_{\text{orange}}$$

$$\frac{\partial L_n}{\partial \beta_1} = \underbrace{\frac{\partial L_n}{\partial \mu_n}}_{\text{blue}} \underbrace{\frac{\partial \mu_n}{\partial \eta_n}}_{\text{yellow}} \underbrace{\frac{\partial \eta_n}{\partial \beta_1}}_{\text{orange}} = \underbrace{\left(\frac{y_n}{\mu_n} - \frac{1 - y_n}{1 - \mu_n} \right)}_{\text{blue}} \times \underbrace{(\mu_n(1 - \mu_n))}_{\text{yellow}} \times \underbrace{x_n}_{\text{orange}}$$

Simplify

$$\frac{\partial L_n}{\partial \beta_0} = \frac{\partial L_n}{\partial \mu_n} \frac{\partial \mu_n}{\partial \eta_n} \frac{\partial \eta_n}{\partial \beta_0} = y_n(1 - \mu_n) - (1 - y_n)\mu_n$$

$$\frac{\partial L_n}{\partial \beta_1} = \frac{\partial L_n}{\partial \mu_n} \frac{\partial \mu_n}{\partial \eta_n} \frac{\partial \eta_n}{\partial \beta_1} = (y_n(1 - \mu_n) - (1 - y_n)\mu_n)x_n$$

Simplify

$$\frac{\partial L_n}{\partial \beta_0} = y_n - y_n \mu_n - \mu_n + y_n \mu_n = y_n - \mu_n$$

$$\frac{\partial L_n}{\partial \beta_1} = (y_n - y_n \mu_n - \mu_n + y_n \mu_n) x_n = (y_n - \mu_n) x_n$$

Finally, the partial derivatives simplify to:

$$\frac{\partial L_n}{\partial \beta_0} = y_n - \mu_n$$

$$\frac{\partial L_n}{\partial \beta_1} = (y_n - \mu_n)x_n$$

Do you see a pattern between the two?

$$\frac{\partial L_n}{\partial \beta_0} = y_n - \mu_n$$

$$\frac{\partial L_n}{\partial \beta_1} = (y_n - \mu_n)x_n$$

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Do you see a pattern between the two?

$$\frac{\partial L_n}{\partial \beta_0} = (y_n - \mu_n) \times 1$$

$$\frac{\partial L_n}{\partial \beta_1} = (y_n - \mu_n) \times x_n$$

Can we write out the derivative with respect to a general β_d -coefficient?

Let's use the same “fake” input trick that we used with linear models

- Write the linear predictor starting from the “zeroth” input:

$$\eta_n = \beta_0 x_{n,0} + \beta_1 x_{n,1}$$

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- Write the linear predictor starting from the “zeroth” input:

$$\eta_n = \beta_0 x_{n,0} + \beta_1 x_{n,1}$$

- Which we can write as a general summation or inner-product:

$$\eta_n = \sum_{d=0}^D (\beta_d x_{n,d}) = \mathbf{x}_{n,:} \boldsymbol{\beta}$$

The partial derivative of the linear predictor with respect to the d -th coefficient is then:

$$\frac{\partial \eta_n}{\partial \beta_d} = \sum_{d'=0}^D \left(\frac{\partial}{\partial \beta_d} (\beta_{d'} x_{n,d'}) \right)$$

- Is non-zero only when $d = d'$:

$$\frac{\partial \eta_n}{\partial \beta_d} = \frac{\partial}{\partial \beta_d} (\beta_d x_{n,d}) = ?$$

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$$\frac{\partial \eta_n}{\partial \beta_d} = x_{n,d}$$

The partial derivative of the n -th likelihood with respect to β_d is then:

$$\frac{\partial L_n}{\partial \beta_d} = (y_n - \mu_n)x_{n,d}$$

Remember that μ_n is a non-linear function of the coefficients!

$$\frac{\partial L_n}{\partial \beta_d} = (y_n - \mu_n)x_{n,d} = (y_n - \text{logit}^{-1}(\eta_n))x_{n,d}$$

Substitute in the definition of the logistic function:

$$\frac{\partial L_n}{\partial \beta_d} = \left(y_n - \frac{\exp(\eta_n)}{1 + \exp(\eta_n)} \right) x_{n,d}$$

Substitute in for the linear predictor:

$$\frac{\partial L_n}{\partial \beta_d} = \left(y_n - \frac{\exp(\mathbf{x}_{n,:}\boldsymbol{\beta})}{1 + \exp(\mathbf{x}_{n,:}\boldsymbol{\beta})} \right) x_{n,d}$$

The derivative of the log-likelihood depends non-linearly on the unknown coefficients!

Substitute in for the linear predictor:

We cannot directly solve for the coefficients!!!

$$\frac{\partial L_n}{\partial \beta_d} = \left(y_n - \frac{\exp(\mathbf{x}_{n,:}\boldsymbol{\beta})}{1 + \exp(\mathbf{x}_{n,:}\boldsymbol{\beta})} \right) x_{n,d}$$

The derivative of the log-likelihood depends non-linearly on the unknown coefficients!

Back to working with the expression in terms of the event probability:

$$\frac{\partial L_n}{\partial \beta_d} = (y_n - \mu_n)x_{n,d}$$

Sum up over all N observations

$$\frac{\partial L}{\partial \beta_d} = \sum_{n=1}^N \left((y_n - \mu_n) x_{n,d} \right)$$

Sum up over all N observations

$$\frac{\partial L}{\partial \beta_d} = \sum_{n=1}^N \left((y_n - \mu_n) x_{n,d} \right)$$

- Rewrite in vector notation:

$$\frac{\partial L}{\partial \beta_d} = \mathbf{x}_{:,d}^T (\mathbf{y} - \boldsymbol{\mu})$$

Substitute back into the gradient vector, for the single input linear relationship

$$\mathbf{g} = \begin{bmatrix} \frac{\partial L}{\partial \beta_0} \\ \frac{\partial L}{\partial \beta_1} \end{bmatrix} = \begin{bmatrix} \mathbf{x}_{:,0}^T (\mathbf{y} - \boldsymbol{\mu}) \\ \mathbf{x}_{:,1}^T (\mathbf{y} - \boldsymbol{\mu}) \end{bmatrix}$$

Substitute back into the gradient vector for the general situation of D inputs

$$\mathbf{g} = \begin{bmatrix} \frac{\partial L}{\partial \beta_{d=0}} \\ \frac{\partial L}{\partial \beta_{d=1}} \\ \vdots \\ \frac{\partial L}{\partial \beta_{d=D}} \end{bmatrix} = \begin{bmatrix} \mathbf{x}_{:,d=0}^T (\mathbf{y} - \boldsymbol{\mu}) \\ \mathbf{x}_{:,d=1}^T (\mathbf{y} - \boldsymbol{\mu}) \\ \vdots \\ \mathbf{x}_{:,d=D}^T (\mathbf{y} - \boldsymbol{\mu}) \end{bmatrix}$$

The gradient vector can be written compactly with the $N \times (D + 1)$ design matrix $\mathbf{X}$

$$\mathbf{g} = \mathbf{X}^T (\mathbf{y} - \boldsymbol{\mu})$$

Contrast with the gradient from the linear model

$$\mathbf{g} = \frac{1}{\sigma^2} (\mathbf{X}^T \mathbf{y} - \mathbf{X}^T \mathbf{X} \boldsymbol{\beta})$$



$$\mathbf{g} = \mathbf{X}^T (\mathbf{y} - \boldsymbol{\mu})$$

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What's this?

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What's this?

Error, residual, difference
between observations and
the mean trend!

Contrast with the gradient from the linear model

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$$\mathbf{g} = \mathbf{X}^T (\mathbf{y} - \boldsymbol{\mu})$$

Difference between
observations and the
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$$\mathbf{g} = \frac{1}{\sigma^2} \mathbf{X}^T (\mathbf{y} - \mathbf{X} \boldsymbol{\beta})$$

What's this?

Error, residual, difference
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Now let's work through deriving the Hessian, the matrix of second derivatives

- Start with terms along the main diagonal of the Hessian matrix
- Second derivative of the n -th log-likelihood with respect to the d -th coefficient:

$$\frac{\partial^2 L_n}{\partial \beta_d^2} = \frac{\partial}{\partial \beta_d} \left(\frac{\partial L_n}{\partial \beta_d} \right)$$

Now let's work through deriving the Hessian, the matrix of second derivatives

- Start with terms along the main diagonal of the Hessian matrix
- Second derivative of the n -th log-likelihood with respect to the d -th coefficient:

$$\frac{\partial^2 L_n}{\partial \beta_d^2} = \frac{\partial}{\partial \beta_d} \left(\frac{\partial L_n}{\partial \beta_d} \right) = \frac{\partial}{\partial \beta_d} \left((y_n - \mu_n) x_{n,d} \right)$$

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$$\frac{\partial^2 L_n}{\partial \beta_d^2} = \frac{\partial}{\partial \beta_d} \left(\frac{\partial L_n}{\partial \beta_d} \right) = \frac{\partial}{\partial \beta_d} \left((y_n - \mu_n) x_{n,d} \right) = -x_{n,d} \frac{\partial \mu_n}{\partial \beta_d}$$

Expand the partial derivative of the event probability via the chain rule

$$\frac{\partial \mu_n}{\partial \beta_d} = \frac{\partial \mu_n}{\partial \eta_n} \times \frac{\partial \eta_n}{\partial \beta_d}$$

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$$\overbrace{\frac{\partial \mu_n}{\partial \eta_n}} = \mu_n(1 - \mu_n)$$

Bernoulli variance!

Expand the partial derivative of the event probability via the chain rule

$$\frac{\partial \mu_n}{\partial \beta_d} = \frac{\partial \mu_n}{\partial \eta_n} \times \boxed{\frac{\partial \eta_n}{\partial \beta_d}}$$

$$\frac{\partial \eta_n}{\partial \beta_d} = x_{n,d}$$

d-th input

Expand the partial derivative of the event probability via the chain rule

$$\frac{\partial \mu_n}{\partial \beta_d} = \mu_n(1 - \mu_n)x_{n,d}$$

Substitute back into the expression for the second derivative

- Second derivative of the n -th log-likelihood with respect to the d -th coefficient:

$$\frac{\partial^2 L_n}{\partial \beta_d^2} = -x_{n,d} \frac{\partial \mu_n}{\partial \beta_d} = -x_{n,d} \mu_n (1 - \mu_n) x_{n,d}$$

Main diagonal terms involve the square of the inputs!

$$\frac{\partial^2 L_n}{\partial \beta_d^2} = -\mu_n(1 - \mu_n)(x_{n,d})^2$$

The off-diagonal terms involve the d -th and d' -th coefficients

$$\frac{\partial^2 L_n}{\partial \beta_{d'} \partial \beta_d} = \frac{\partial}{\partial \beta_{d'}} \left(\frac{\partial L_n}{\partial \beta_d} \right)$$

The off-diagonal terms involve the d -th and d' -th coefficients

$$\frac{\partial^2 L_n}{\partial \beta_{d'} \partial \beta_d} = \frac{\partial}{\partial \beta_{d'}} \left(\frac{\partial L_n}{\partial \beta_d} \right) = \frac{\partial}{\partial \beta_{d'}} \left((y_n - \mu_n) x_{n,d} \right)$$

The off-diagonal terms involve the d -th and d' -th coefficients

$$\frac{\partial^2 L_n}{\partial \beta_{d'} \partial \beta_d} = \frac{\partial}{\partial \beta_{d'}} \left(\frac{\partial L_n}{\partial \beta_d} \right) = \frac{\partial}{\partial \beta_{d'}} \left((y_n - \mu_n) x_{n,d} \right) = -x_{n,d} \frac{\partial \mu_n}{\partial \beta_{d'}}$$

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$$\frac{\partial^2 L_n}{\partial \beta_{d'} \partial \beta_d} = \frac{\partial}{\partial \beta_{d'}} \left(\frac{\partial L_n}{\partial \beta_d} \right) = \frac{\partial}{\partial \beta_{d'}} \left((y_n - \mu_n) x_{n,d} \right) = -x_{n,d} \frac{\partial \mu_n}{\partial \beta_{d'}}$$

We just went through this derivation! The notation is just slightly changed.

Follow the same steps to expand with the chain rule

$$\frac{\partial \mu_n}{\partial \beta_{d'}} = \frac{\partial \mu_n}{\partial \eta_n} \times \frac{\partial \eta_n}{\partial \beta_{d'}} = \mu_n(1 - \mu_n)x_{n,d'}$$

The off-diagonal terms involve the d -th and d' -th coefficients

$$\frac{\partial^2 L_n}{\partial \beta_{d'} \partial \beta_d} = -x_{n,d} \frac{\partial \mu_n}{\partial \beta_{d'}} = -x_{n,d} \mu_n (1 - \mu_n) x_{n,d'}$$

Off-diagonal terms involve multiplying the **pair wise** products of the inputs!

Need to sum over all N observations

- Main diagonal terms:

$$\frac{\partial^2 L}{\partial \beta_d^2} = \sum_{n=1}^N \left(\frac{\partial^2 L_n}{\partial \beta_d^2} \right) = \sum_{n=1}^N \left(-\mu_n (1 - \mu_n) (x_{n,d})^2 \right)$$

- Off-diagonal terms:

$$\frac{\partial^2 L}{\partial \beta_{d'} \partial \beta_d} = \sum_{n=1}^N \left(\frac{\partial^2 L_n}{\partial \beta_{d'} \partial \beta_d} \right) = \sum_{n=1}^N \left(-\mu_n (1 - \mu_n) x_{n,d} x_{n,d'} \right)$$

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Sum over all squared input values

- Off-diagonal terms:

$$\frac{\partial^2 L}{\partial \beta_{d'} \partial \beta_d} = \sum_{n=1}^N \left(\frac{\partial^2 L_n}{\partial \beta_{d'} \partial \beta_d} \right) = \sum_{n=1}^N \left(-\mu_n (1 - \mu_n) x_{n,d} x_{n,d'} \right)$$

Sum over all pair-wise input products

Analogous to the sum-of-squares matrix from the linear model

- With logistic regression the sum-of-squares is **WEIGHTED** based on the Bernoulli variance!

$$\mathbf{H} = - \sum_{n=1}^N (\mu_n (1 - \mu_n) \mathbf{x}_{n,:}^T \mathbf{x}_{n,:})$$

- The weights, $\mu_n (1 - \mu_n)$, vary across observations.

Compactly write the Hessian in terms of the matrix sum-of-squares and a weighting matrix

- The weighting matrix **S** is a diagonal matrix of Bernoulli weights:

$$\mathbf{S} = \text{diag}\{\mu_n(1 - \mu_n)\}_{n=1}^N$$

S is $N \times N$

The weighting matrix is a diagonal matrix, the only non-zero terms are along the main diagonal

$$\mathbf{S} = \begin{bmatrix} \mu_1(1 - \mu_1) & 0 & \cdots & 0 & 0 \\ 0 & \mu_2(1 - \mu_2) & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & \mu_{N-1}(1 - \mu_{N-1}) & 0 \\ 0 & 0 & \cdots & 0 & \mu_N(1 - \mu_N) \end{bmatrix}$$

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- The weighting matrix $\mathbf{S}$ is a diagonal matrix of Bernoulli weights:

$$\mathbf{S} = \text{diag}\{\mu_n(1 - \mu_n)\}_{n=1}^N$$

- Hessian matrix:

$$\mathbf{H} = -\mathbf{X}^T \mathbf{S} \mathbf{X}$$

In linear models, the sum-of-squares matrix added together contributions from individual observations

- $\mathbf{X}^T \mathbf{X}$ based on the first 3 observations:

$$\mathbf{X}^T \mathbf{X} = \sum_{n=1}^{N=3} (\mathbf{x}_{n,:}^T \mathbf{x}_{n,:}) = \mathbf{x}_{1,:}^T \mathbf{x}_{1,:} + \mathbf{x}_{2,:}^T \mathbf{x}_{2,:} + \mathbf{x}_{3,:}^T \mathbf{x}_{3,:}$$

- With a linear relationship with a single input $D = 1$: $\mu_n = \beta_0 + \beta_1 x_{n,1}$
- Each outer product is 2×2 :

$$\mathbf{X}^T \mathbf{X} = \sum_{n=1}^{N=3} (\mathbf{x}_{n,:}^T \mathbf{x}_{n,:}) = \begin{bmatrix} 1 & x_{1,1} \\ x_{1,1} & x_{1,1}^2 \end{bmatrix} + \begin{bmatrix} 1 & x_{2,1} \\ x_{2,1} & x_{2,1}^2 \end{bmatrix} + \begin{bmatrix} 1 & x_{3,1} \\ x_{3,1} & x_{3,1}^2 \end{bmatrix}$$

In logistic regression, the weighted sum-of-squares matrix multiplies each observation's contribution by the variance of the outcome!

- $\mathbf{X}^T \mathbf{S} \mathbf{X}$ based on the first 3 observations:

$$\mathbf{X}^T \mathbf{S} \mathbf{X} = \sum_{n=1}^{N=3} (\mu_n (1 - \mu_n) \mathbf{x}_{n,:}^T \mathbf{x}_{n,:})$$

- Each $\mu_n (1 - \mu_n)$ is a scalar value.

$$\mathbf{X}^T \mathbf{S} \mathbf{X} = \mu_1 (1 - \mu_1) \mathbf{x}_{1,:}^T \mathbf{x}_{1,:} + \mu_2 (1 - \mu_2) \mathbf{x}_{2,:}^T \mathbf{x}_{2,:} + \mu_3 (1 - \mu_3) \mathbf{x}_{3,:}^T \mathbf{x}_{3,:}$$

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- For a single input with a linear relationship, $\eta_n = \beta_0 + \beta_1 x_{n,1}$, the outer products are still 2×2 !

$$\mu_1(1 - \mu_1) \begin{bmatrix} 1 & x_{1,1} \\ x_{1,1} & x_{1,1}^2 \end{bmatrix} + \mu_2(1 - \mu_2) \begin{bmatrix} 1 & x_{2,1} \\ x_{2,1} & x_{2,1}^2 \end{bmatrix} + \mu_3(1 - \mu_3) \begin{bmatrix} 1 & x_{3,1} \\ x_{3,1} & x_{3,1}^2 \end{bmatrix}$$

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If the event probabilities are $\mu_1 = 0.5$, $\mu_2 = 0.15$, and $\mu_3 = 0.85$ which observation will have the highest weighted contribution?

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$$0.5 \times (1 - 0.5) \begin{bmatrix} 1 & x_{1,1} \\ x_{1,1} & x_{1,1}^2 \end{bmatrix} + 0.15 \times (1 - 0.15) \begin{bmatrix} 1 & x_{2,1} \\ x_{2,1} & x_{2,1}^2 \end{bmatrix} + 0.85 \times (1 - 0.85) \begin{bmatrix} 1 & x_{3,1} \\ x_{3,1} & x_{3,1}^2 \end{bmatrix}$$

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- **Observation 1 has the highest weight!** Observations 2 and 3 have the same weight because the Bernoulli variances are equal.

$$0.25 \times \begin{bmatrix} 1 & x_{1,1} \\ x_{1,1} & x_{1,1}^2 \end{bmatrix} + 0.1275 \times \begin{bmatrix} 1 & x_{2,1} \\ x_{2,1} & x_{2,1}^2 \end{bmatrix} + 0.1275 \times \begin{bmatrix} 1 & x_{3,1} \\ x_{3,1} & x_{3,1}^2 \end{bmatrix}$$

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You can view the weighting matrix as reducing the “effective sample size” because we will never add a “full” contribution of each observation to the weighted sum-of-squares!

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This is why logistic regression requires **MORE** observations than the linear model to achieve the same posterior precision in the unknown coefficients.

$[\tilde{\mu}_{1,1} \quad \tilde{\mu}_{1,1}]$

$[\tilde{\mu}_{2,1} \quad \tilde{\mu}_{2,1}]$

$[\tilde{\mu}_{3,1} \quad \tilde{\mu}_{3,1}]$

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This is also why linear separation is a problem.
If the event probability is 0 or 1, the Bernoulli variance for that observation, $\mu(1 - \mu)$, is **ZERO**!

$[\tilde{\mu}_{1,1} \quad \tilde{\mu}_{1,1}]$

$[\tilde{\mu}_{2,1} \quad \tilde{\mu}_{2,1}]$

$[\tilde{\mu}_{3,1} \quad \tilde{\mu}_{3,1}]$

You can view the weighting matrix as reducing the “effective sample size” because we will never add a “full” contribution of each observation to the weighted sum-of-squares!

We saw that the unknowns are NON-LINEARLY related to the log-likelihood

- Thus, we cannot directly solve for the β 's like we did with linear models!
- The OLS formula cannot be used to calculate the MLE of the coefficients!
- We must use an optimization scheme to find the mode!

Use the gradient and Hessian matrix to iteratively update the coefficients

$$\mathbf{g} = \mathbf{X}^T (\mathbf{y} - \boldsymbol{\mu})$$

$$\mathbf{H} = -\mathbf{X}^T \mathbf{S} \mathbf{X}$$

For a given guess of the $\boldsymbol{\beta}$ -parameters, $\boldsymbol{\beta}_k$:

- Evaluate the linear predictor, $\boldsymbol{\eta}_k$
- Evaluate the probability of the event, $\boldsymbol{\mu}_k$
- Calculate the gradient $\mathbf{g}_k$
- Calculate the Bernoulli variance weighting matrix $\mathbf{S}_k$
- Calculate the Hessian matrix $\mathbf{H}_k$
- **Update coefficients based on the gradient and Hessian matrix using the Newton method!**

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Iteratively Reweighted
Least Squares (IRLS)

The $k + 1$ iteration updated “guess” to the coefficients using the “full” Newton method

$$\boldsymbol{\beta}_{k+1} = \boldsymbol{\beta}_k - \mathbf{H}_k^{-1} \mathbf{g}_k$$

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$$\boldsymbol{\beta}_{k+1} = \boldsymbol{\beta}_k - \mathbf{H}_k^{-1} \mathbf{g}_k$$

- Plug in the expression for the gradient:

$$\boldsymbol{\beta}_{k+1} = \boldsymbol{\beta}_k - \mathbf{H}_k^{-1} \mathbf{X}^T (\mathbf{y} - \boldsymbol{\mu}_k)$$

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- Plug in the expression for the gradient:

$$\boldsymbol{\beta}_{k+1} = \boldsymbol{\beta}_k - \mathbf{H}_k^{-1} \mathbf{X}^T (\mathbf{y} - \boldsymbol{\mu}_k)$$

- Plug in the expression for the Hessian:

$$\boldsymbol{\beta}_{k+1} = \boldsymbol{\beta}_k + (\mathbf{X}^T \mathbf{S}_k \mathbf{X})^{-1} \mathbf{X}^T (\mathbf{y} - \boldsymbol{\mu}_k)$$

Perform some algebra

$$\boldsymbol{\beta}_{k+1} = (\mathbf{X}^T \mathbf{S}_k \mathbf{X})^{-1} [(\mathbf{X}^T \mathbf{S}_k \mathbf{X}) \boldsymbol{\beta}_k + \mathbf{X}^T (\mathbf{y} - \boldsymbol{\mu}_k)]$$

Perform some algebra

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$$\boldsymbol{\beta}_{k+1} = (\mathbf{X}^T \mathbf{S}_k \mathbf{X})^{-1} \mathbf{X}^T [\mathbf{S}_k \mathbf{X} \boldsymbol{\beta}_k + \mathbf{y} - \boldsymbol{\mu}_k]$$

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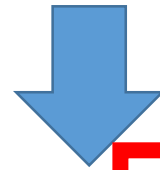
$$\boldsymbol{\beta}_{k+1} = (\mathbf{X}^T \mathbf{S}_k \mathbf{X})^{-1} \mathbf{X}^T \mathbf{S}_k [\mathbf{X} \boldsymbol{\beta}_k + \mathbf{S}_k^{-1} (\mathbf{y} - \boldsymbol{\mu}_k)]$$

Perform some algebra

$$\boldsymbol{\beta}_{k+1} = (\mathbf{X}^T \mathbf{S}_k \mathbf{X})^{-1} [(\mathbf{X}^T \mathbf{S}_k \mathbf{X}) \boldsymbol{\beta}_k + \mathbf{X}^T (\mathbf{y} - \boldsymbol{\mu}_k)]$$



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The “working response” for the k -th iteration

$$\mathbf{z}_k = \mathbf{X}\boldsymbol{\beta}_k + \mathbf{S}_k^{-1}(\mathbf{y} - \boldsymbol{\mu}_k)$$

The “working response” for the k -th iteration

$$\mathbf{z}_k = \mathbf{X}\boldsymbol{\beta}_k + \mathbf{S}_k^{-1}(\mathbf{y} - \boldsymbol{\mu}_k)$$

- Plug back into the $k + 1$ update:

$$\boldsymbol{\beta}_{k+1} = (\mathbf{X}^T \mathbf{S}_k \mathbf{X})^{-1} \mathbf{X}^T \mathbf{S}_k \mathbf{z}_k$$

Compare the IRLS update to the OLS formula from standard linear models

$$\boldsymbol{\beta}_{k+1} = (\mathbf{X}^T \mathbf{S}_k \mathbf{X})^{-1} \mathbf{X}^T \mathbf{S}_k \mathbf{z}_k \qquad \hat{\boldsymbol{\beta}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$$

- The structure or format of the IRLS update is very similar to the OLS formula!
- The IRLS update is a WEIGHTED Least Squares problem.

Compare the IRLS update to the OLS formula from standard linear models

$$\boldsymbol{\beta}_{k+1} = (\mathbf{X}^T \mathbf{S}_k \mathbf{X})^{-1} \mathbf{X}^T \mathbf{S}_k \mathbf{z}_k \qquad \hat{\boldsymbol{\beta}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$$

- At each iteration, the “working responses” are weighted by the Bernoulli variances.
- Highest Bernoulli variance occurs when $\mu = 0.5$, so those observations influence the parameters the most!
- The weights depend on the coefficients, so the **weights change across iterations**.

Take a closer look at the working response

$$\mathbf{z}_k = \underbrace{\mathbf{X}\boldsymbol{\beta}_k}_{\text{What's this?}} + \mathbf{S}_k^{-1}(\mathbf{y} - \boldsymbol{\mu}_k)$$

What's this?

Take a closer look at the working response

$$\mathbf{z}_k = \underbrace{\mathbf{X}\boldsymbol{\beta}_k}_{\text{linear predictor}} + \mathbf{S}_k^{-1}(\mathbf{y} - \boldsymbol{\mu}_k)$$

The k -th iteration's
linear predictor $\boldsymbol{\eta}_k$!!

The working response is an adjusted linear predictor.
How is it adjusted?

The weighting matrix $\mathbf{S}$ is diagonal

- Each element along the main diagonal is the Bernoulli variance associated with the observation. The k -th iteration's weighting matrix is:

$$\mathbf{S}_k = \text{diag}\{\mu_{kn}(1 - \mu_{kn})\}_{n=1}^N$$

- Since it is diagonal, the inverse is also diagonal with elements equal to the inverses of the main diagonal elements:

$$\mathbf{S}_k^{-1} = \text{diag}\left\{\frac{1}{\mu_{kn}(1 - \mu_{kn})}\right\}_{n=1}^N$$

The n -th observation's working response at the k -th iteration, z_{kn}

$$z_{kn} = \mathbf{x}_{n,:} \boldsymbol{\beta}_k + \frac{y_n - \mu_{kn}}{\mu_{kn}(1 - \mu_{kn})}$$

The n -th observation's working response at the k -th iteration, z_{kn}

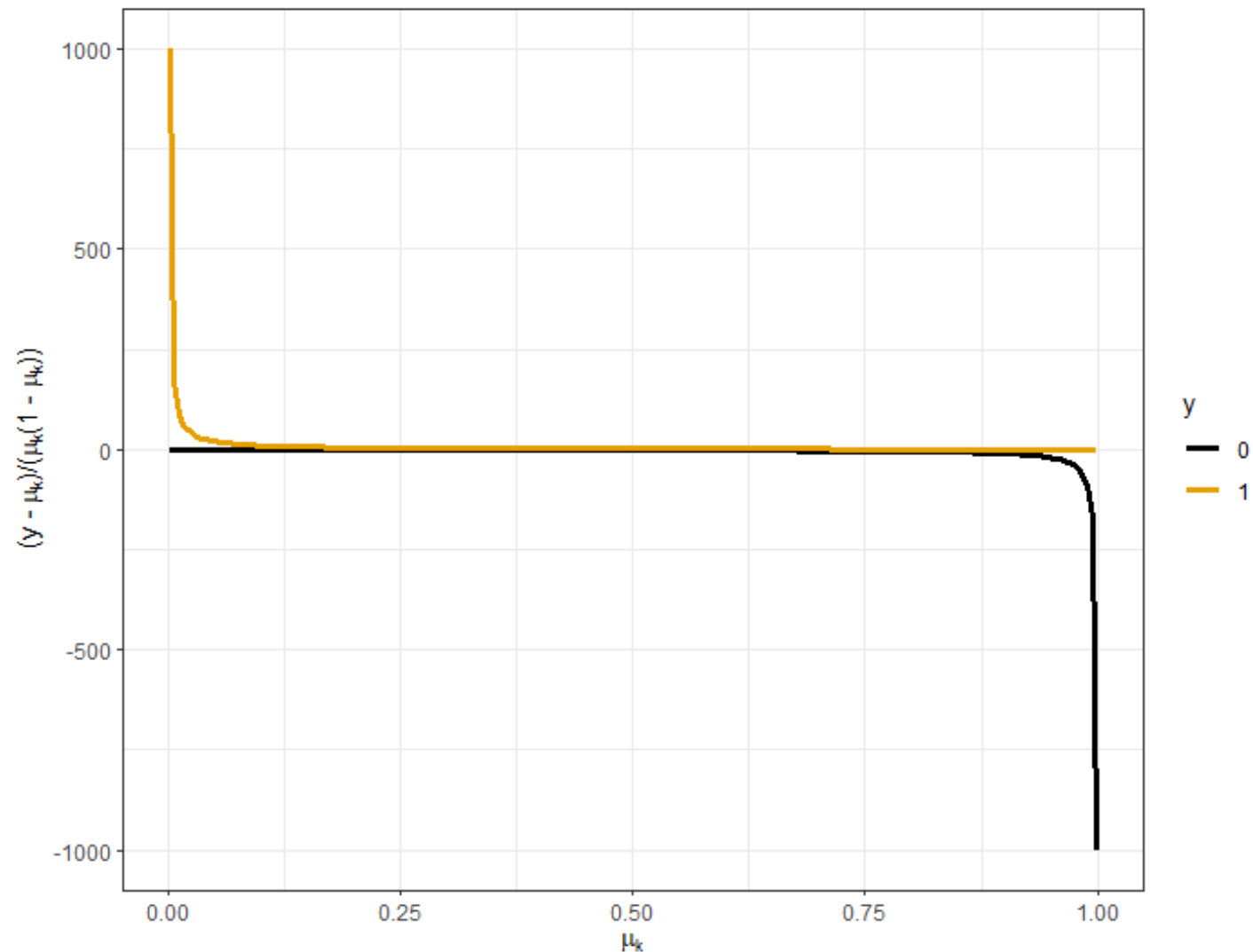
The response y_n is either 0 or 1.

The probability μ_{kn} is between 0 or 1.

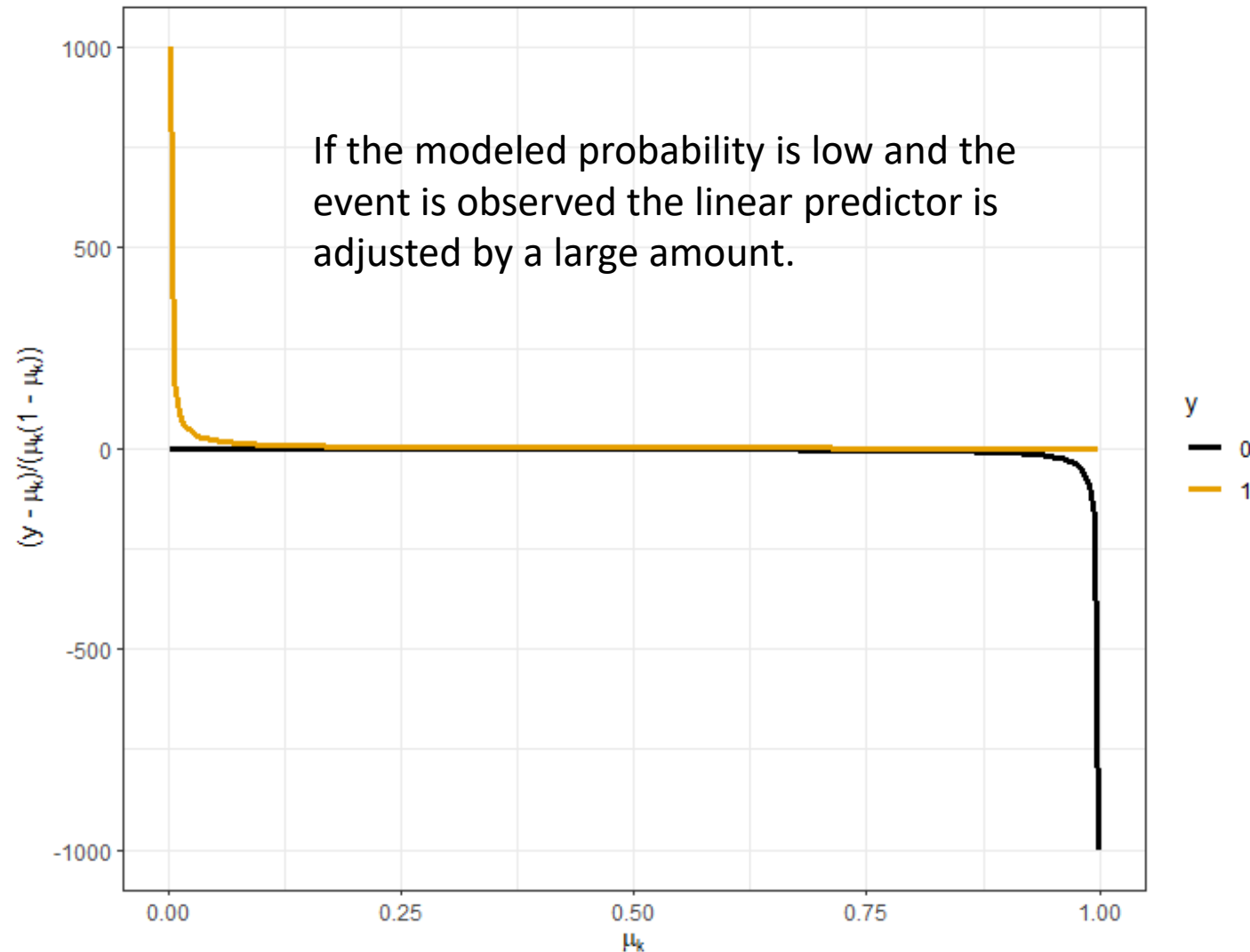
$$z_{kn} = \mathbf{x}_{n,:} \boldsymbol{\beta}_k + \frac{y_n - \mu_{kn}}{\mu_{kn}(1 - \mu_{kn})}$$

How does this “adjustment term” behave?

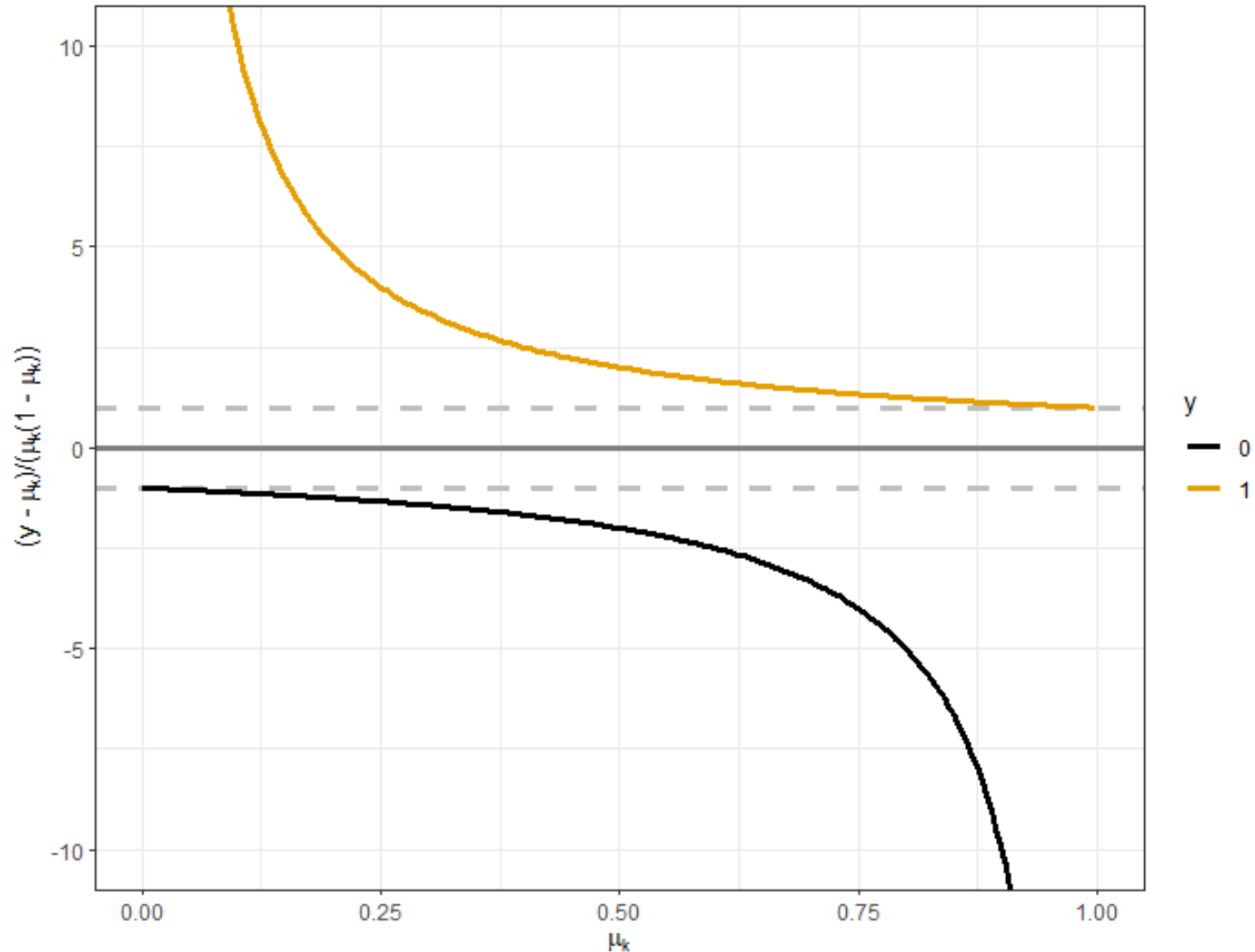
Working response adjustment term blows up when the probability is “inconsistent” with the response



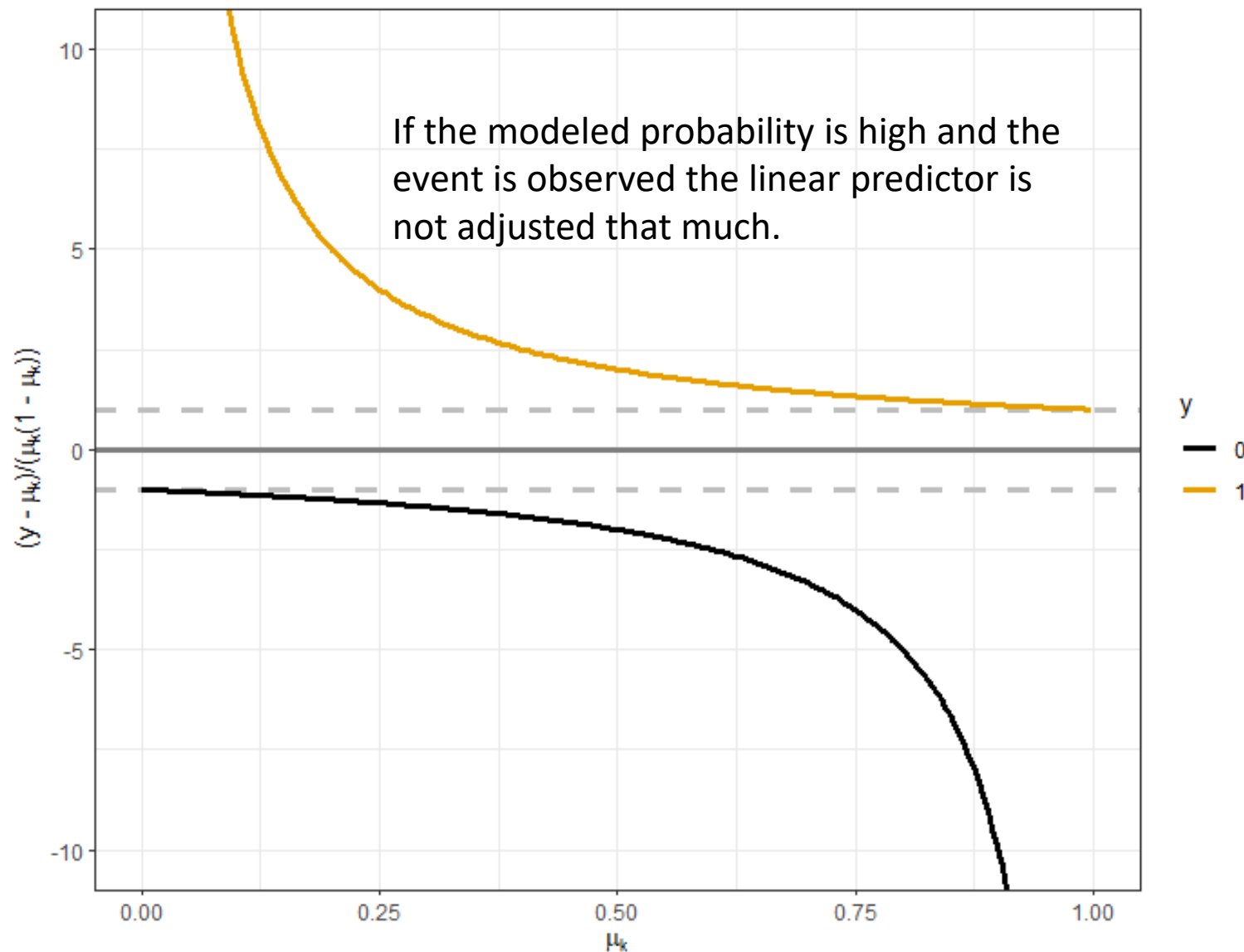
Working response adjustment term blows up when the probability is “inconsistent” with the response



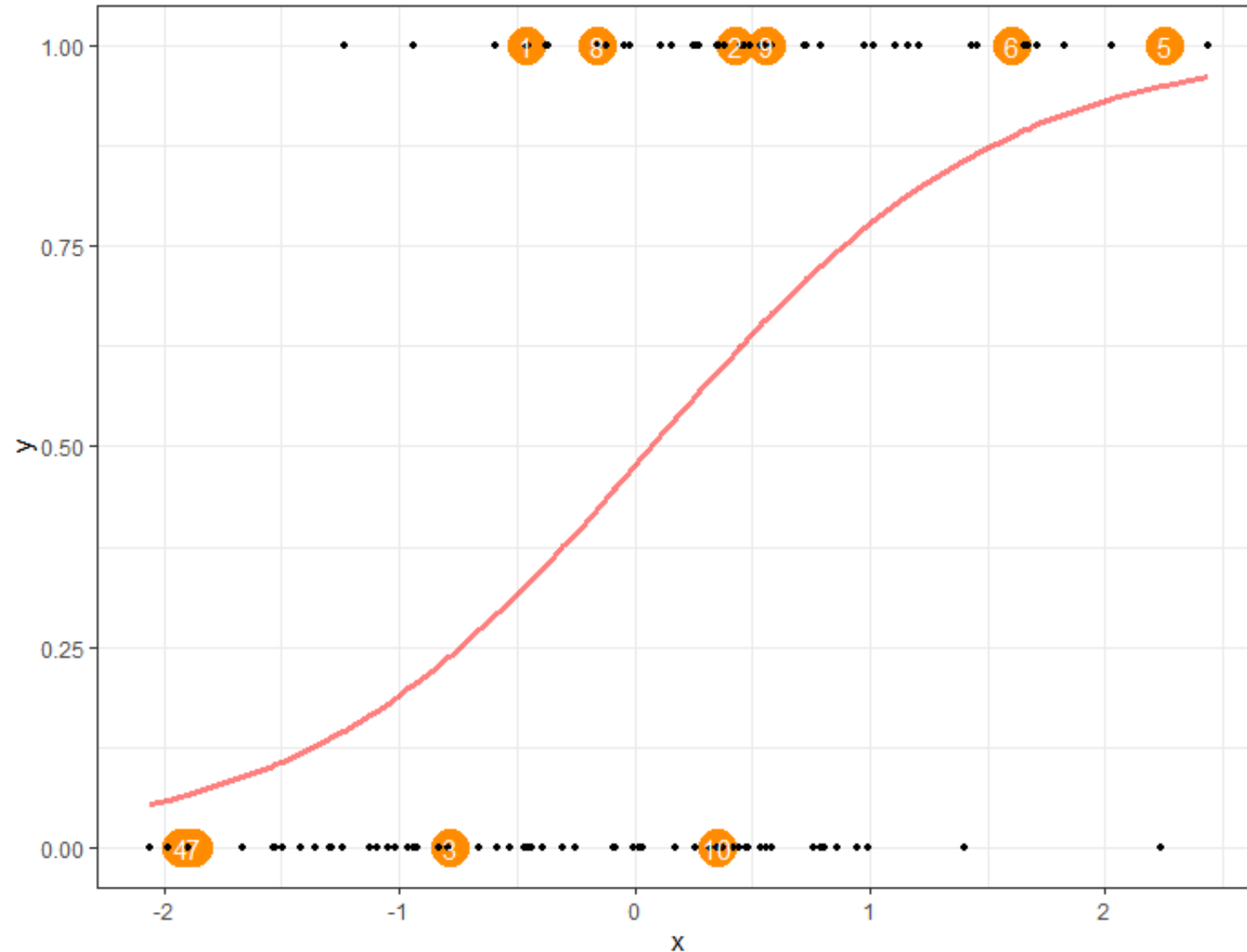
Zoom in: when the probability is “consistent” with the response, the adjustment converges to 1 or -1



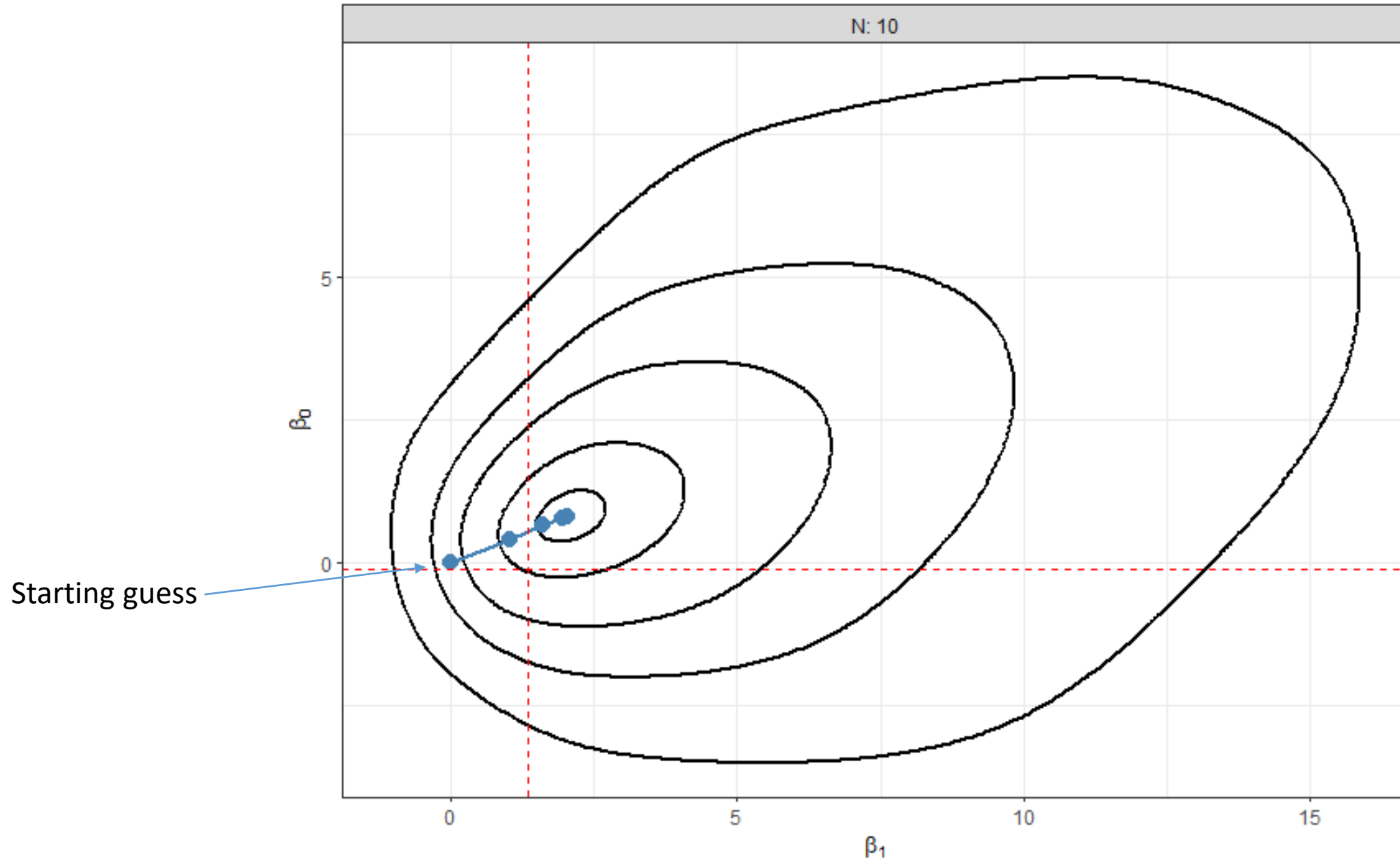
Zoom in: when the probability is “consistent” with the response, the adjustment converges to 1 or -1

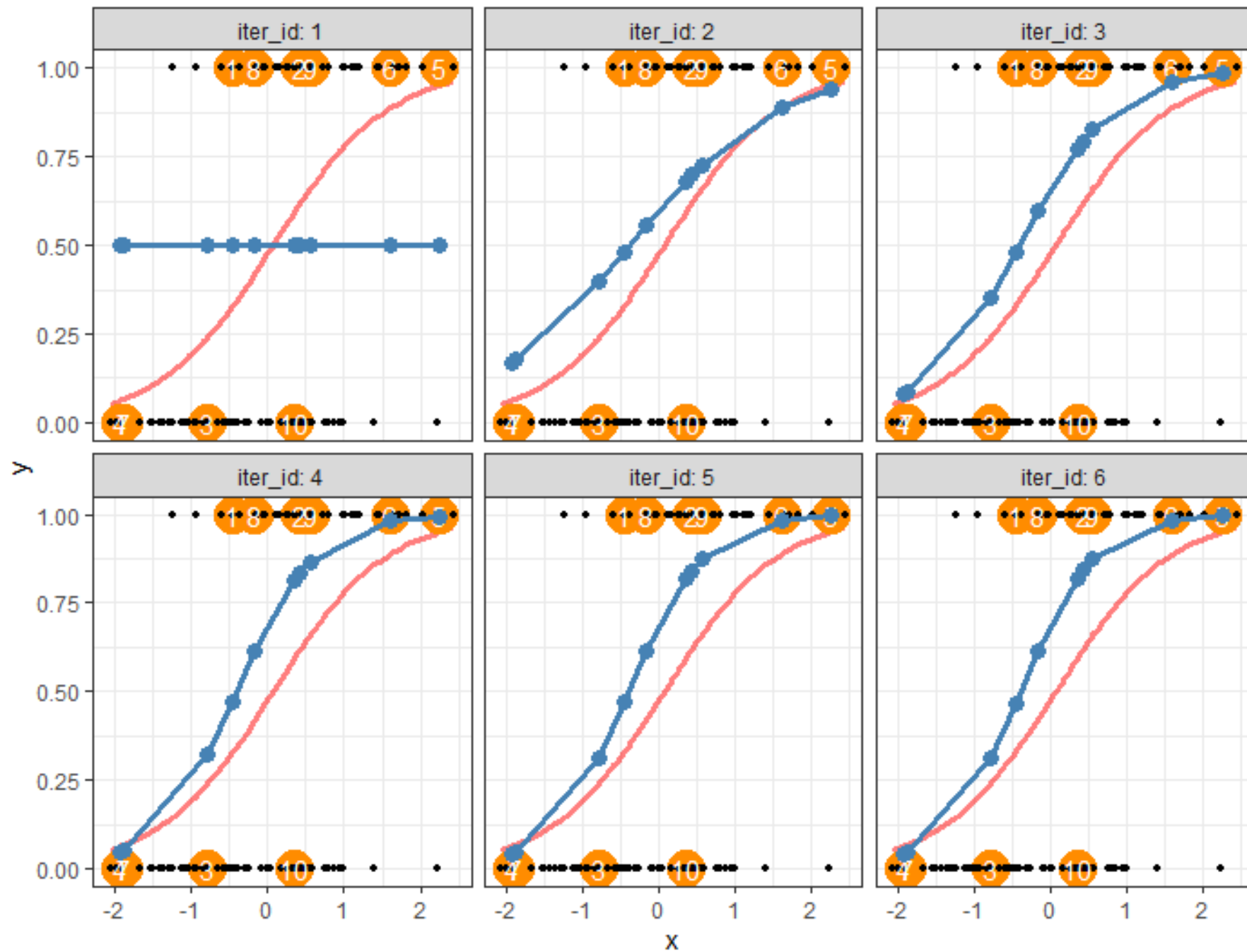


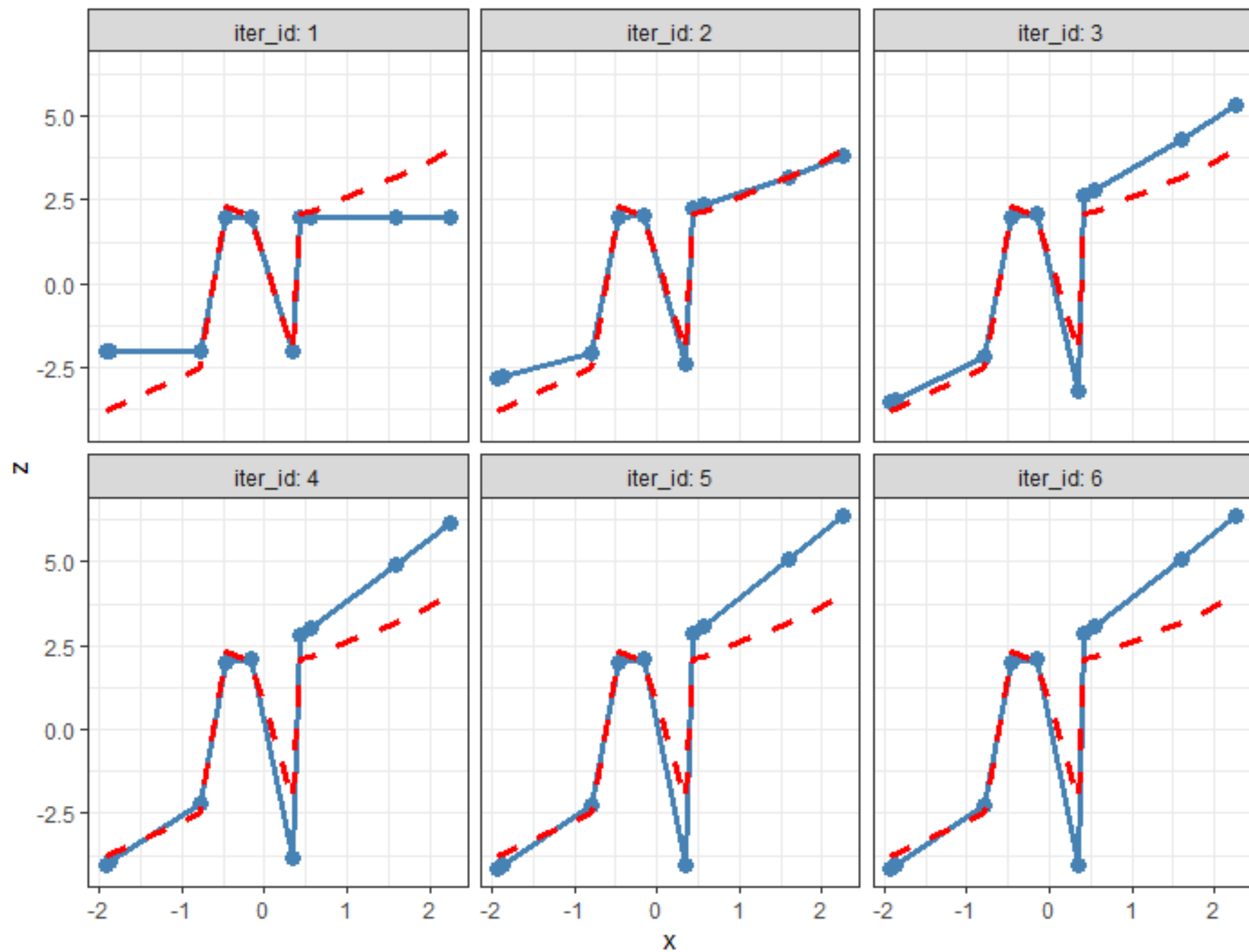
Apply IRLS to the our synthetic data demo with 10 observations



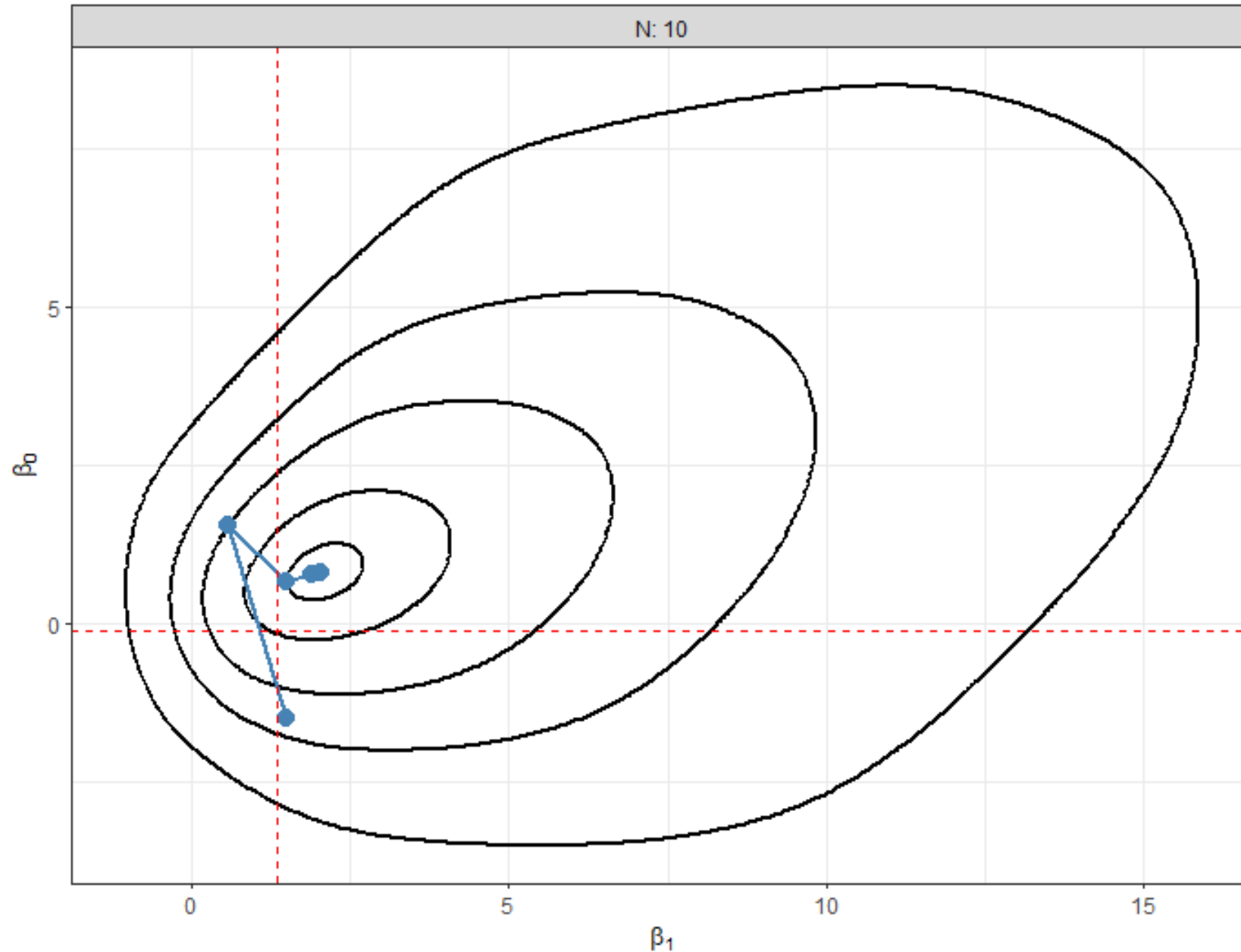
Use a starting guess of 0 for the intercept and the slope

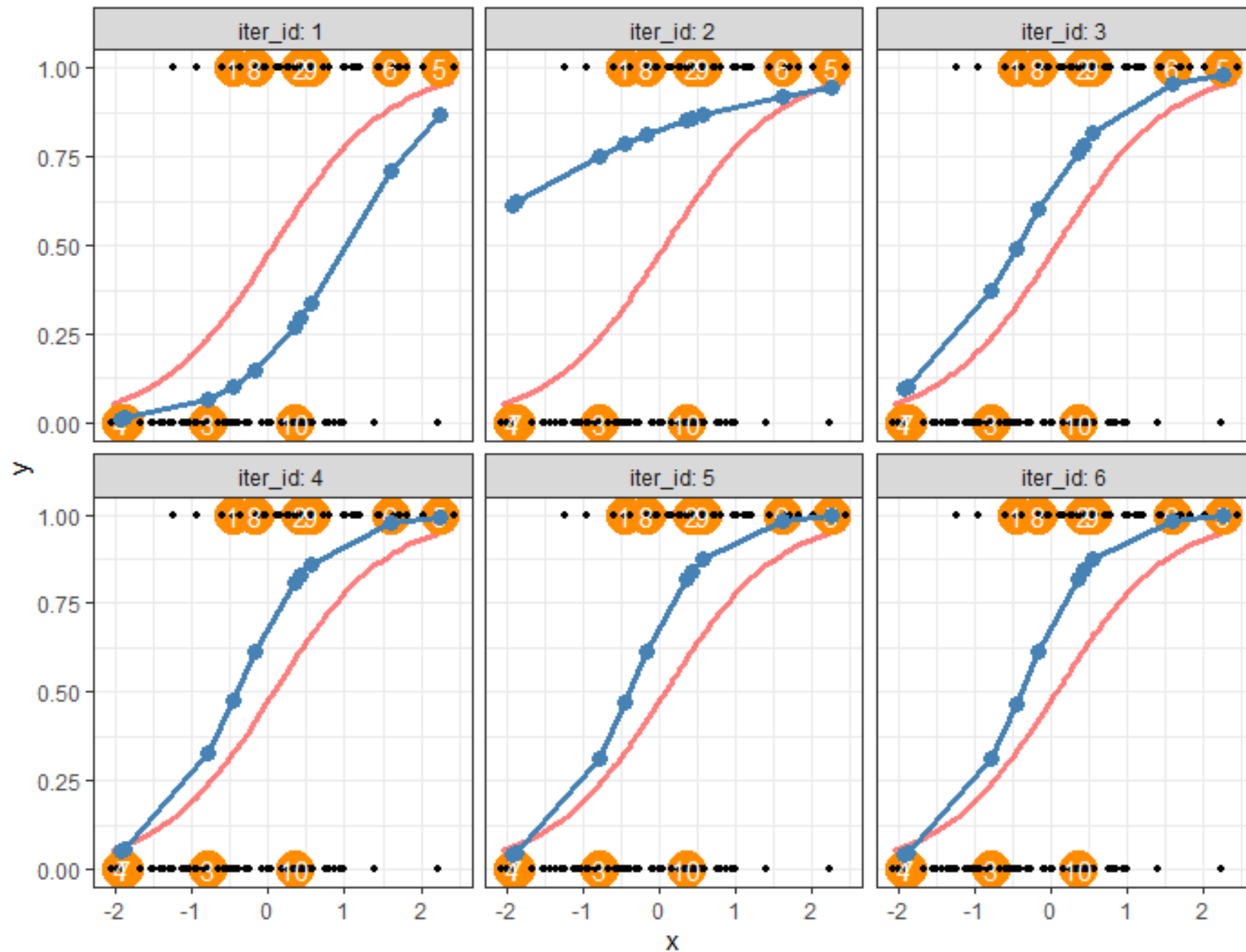


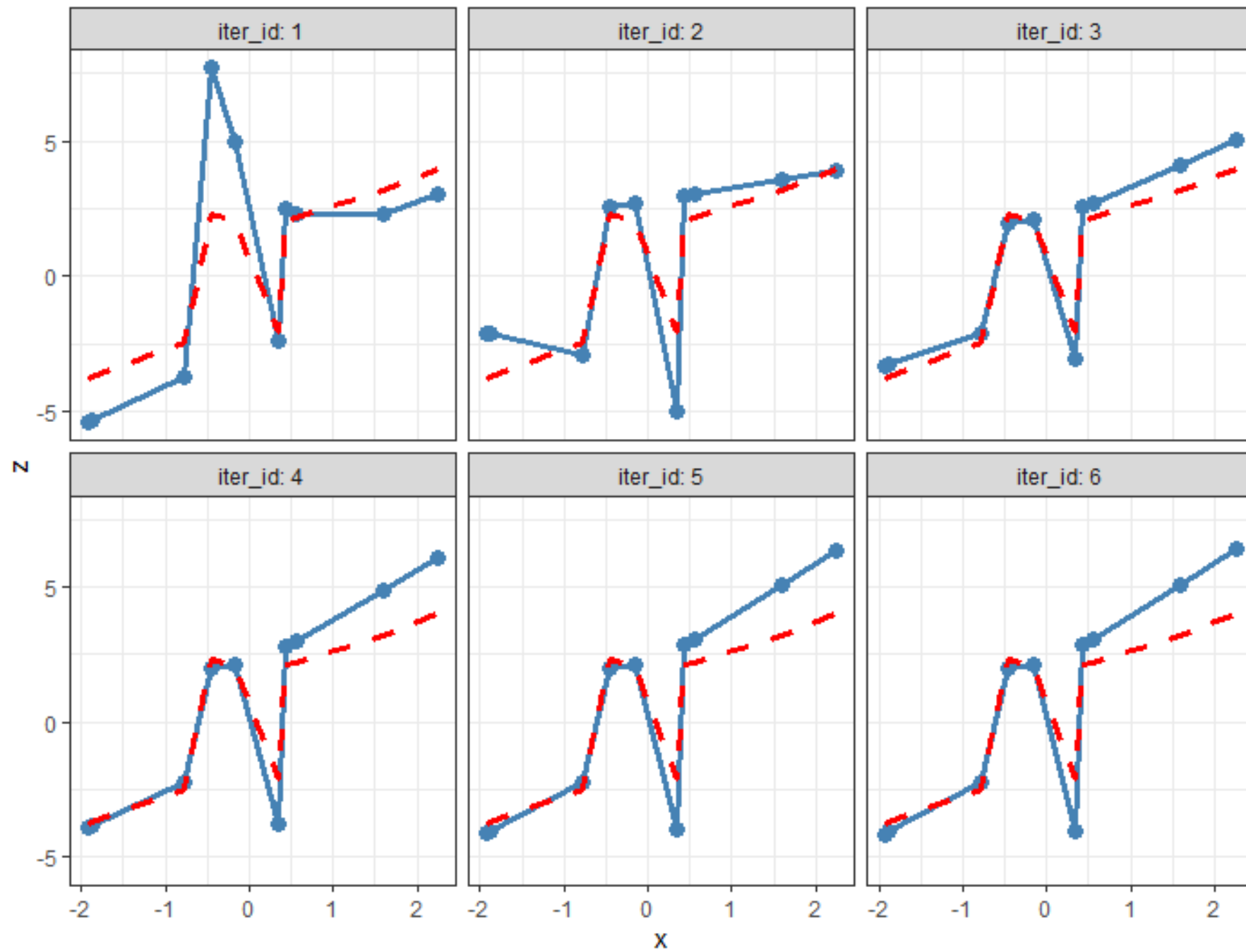




Try another starting guess of $\beta_0 = -1.5$ and $\beta_1 = 1.5$







In the Bayesian setting with an infinitely diffuse prior, to complete the Laplace Approximation...

- Assume we found the posterior mode, $\hat{\boldsymbol{\beta}}$, through an iterative optimization scheme.
- The approximate posterior covariance matrix is:

$$\text{cov}(\boldsymbol{\beta}, \boldsymbol{\beta}) \approx - \left[\mathbf{H} \Big|_{\hat{\boldsymbol{\beta}}} \right]^{-1} = (\mathbf{X}^T \mathbf{S}_{\hat{\boldsymbol{\beta}}} \mathbf{X})^{-1}$$

In the Bayesian setting with an infinitely diffuse prior, to complete the Laplace Approximation...

- Assume we found the posterior mode, $\hat{\boldsymbol{\beta}}$, through an iterative optimization scheme.
- The MVN approximate posterior distribution is:

$$p(\boldsymbol{\beta} \mid \mathbf{y}, \mathbf{X}) \approx \mathcal{N} \left(\boldsymbol{\beta} \mid \hat{\boldsymbol{\beta}}, (\mathbf{X}^T \mathbf{S}_{\hat{\boldsymbol{\beta}}} \mathbf{X})^{-1} \right)$$

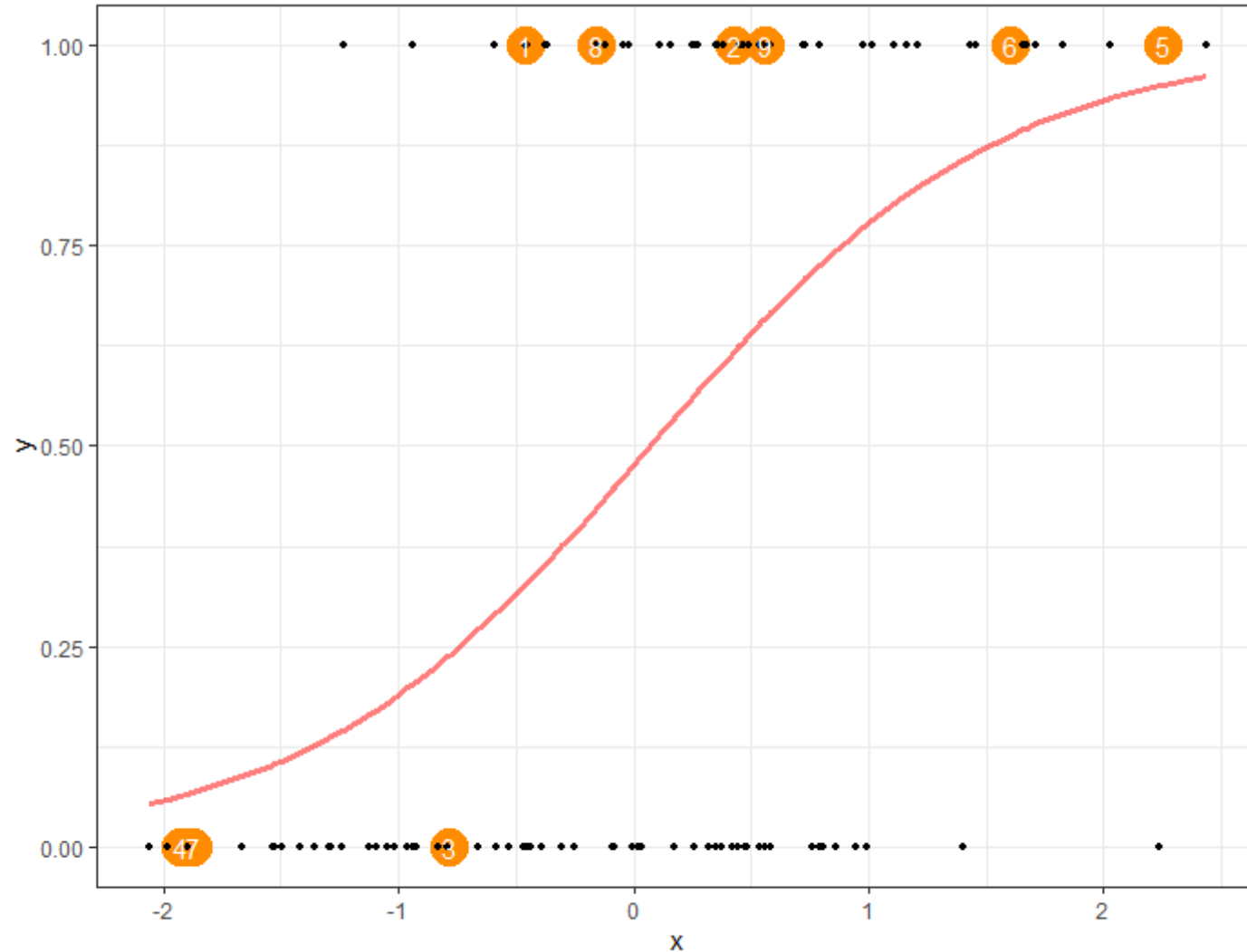
When we have an informative prior

- The derived equations on the gradient and Hessian are not complete.
- Those expressions are missing the influence of the prior!
- However, the key concepts around the influence of the Bernoulli variance on the results are still valid!

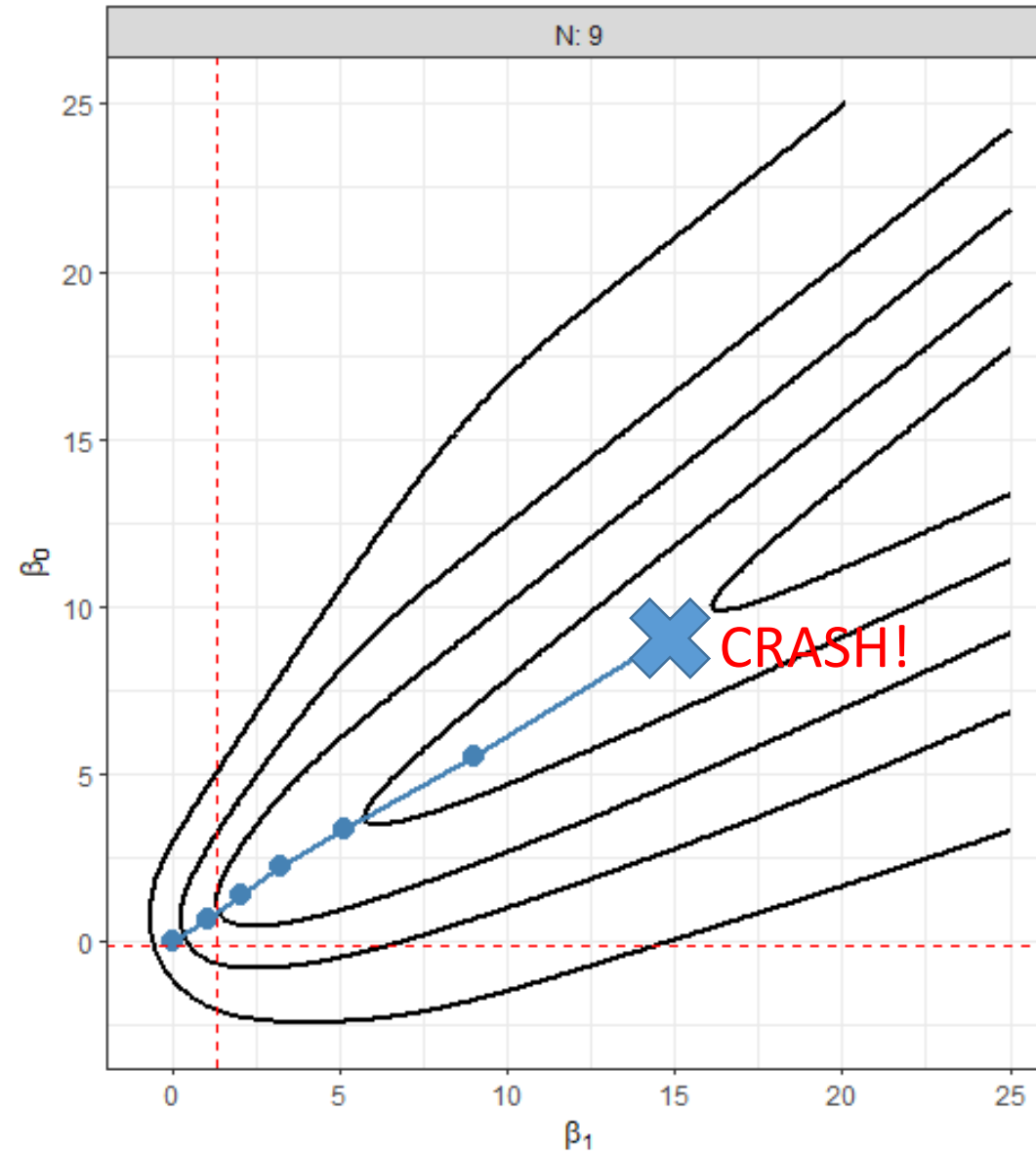
What does the prior provide? Why should we use a prior?

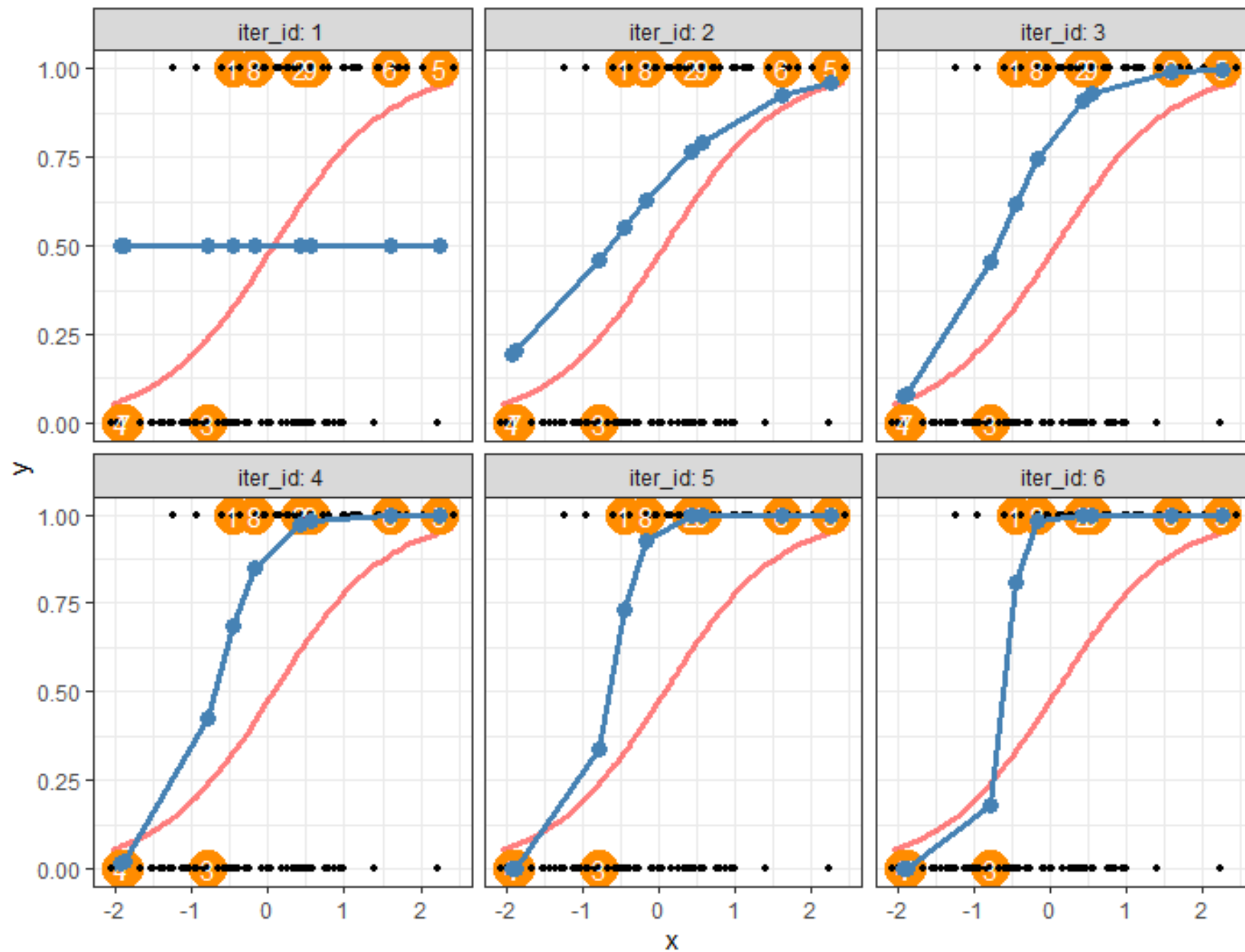
- Let's use just the first 9 observations, instead of the first 10.
- In this example, the two classes are linearly separated when we use the first 9 observations.

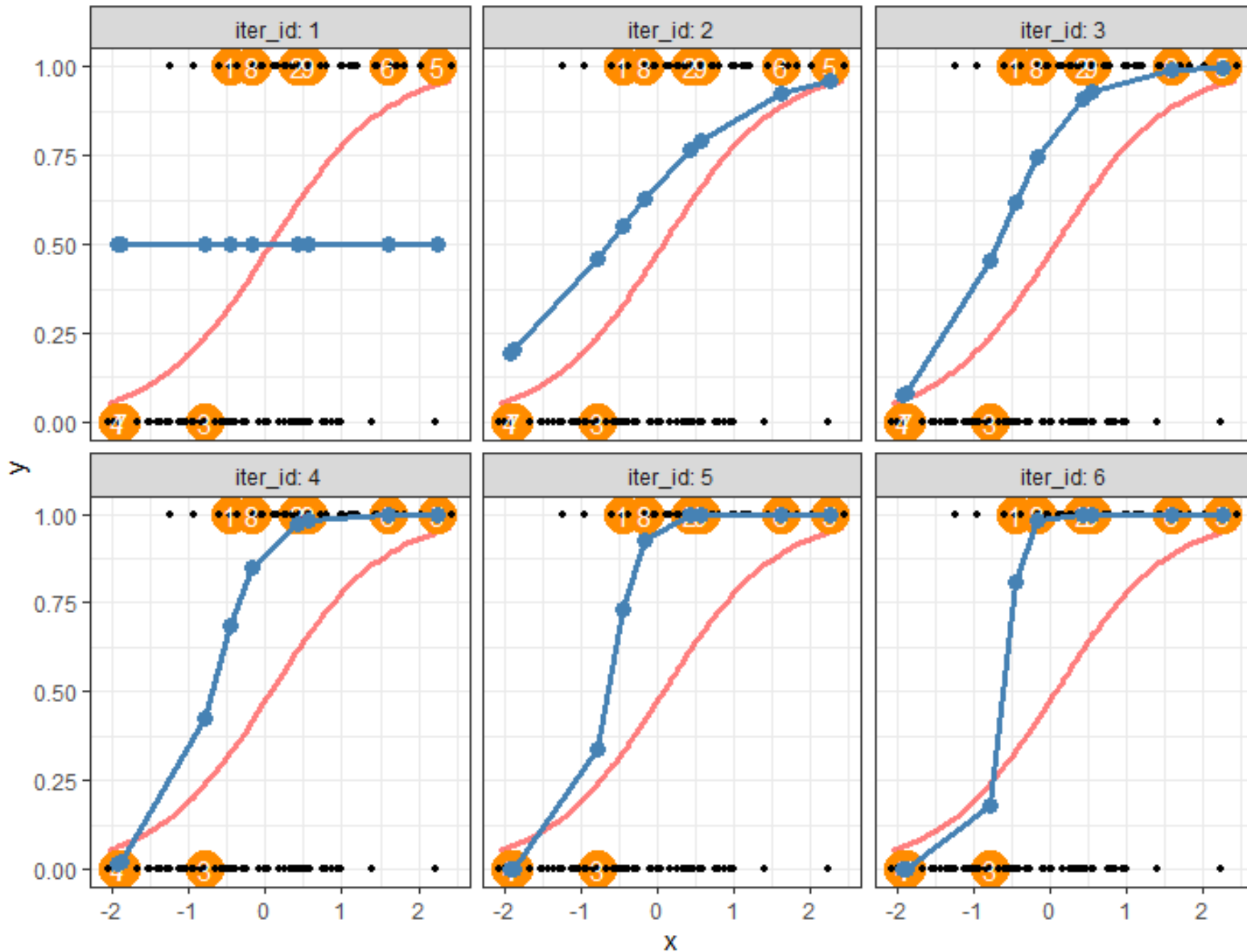
Can draw a separating line between observation 1 and observation 3



The IRLS algorithm will crash!

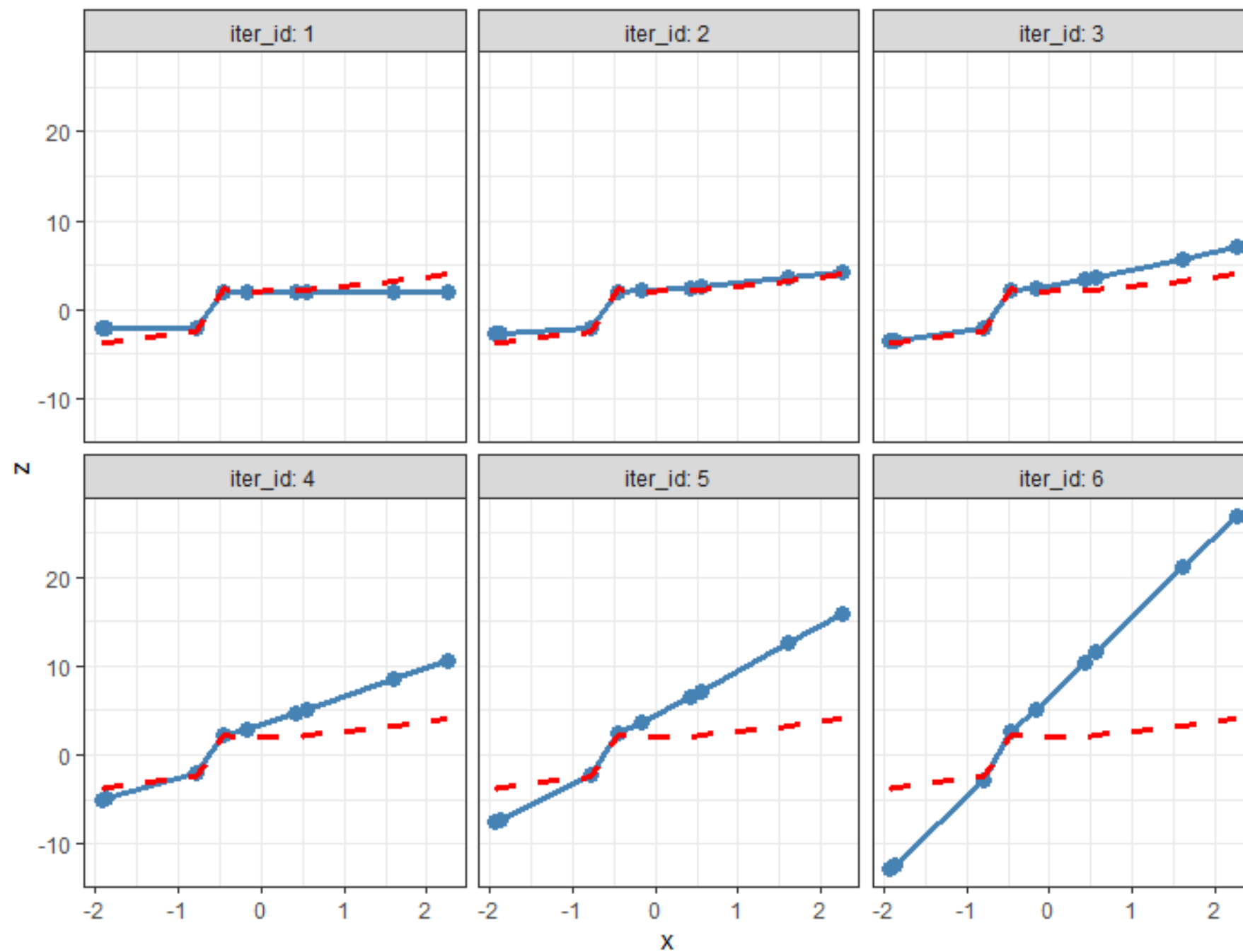


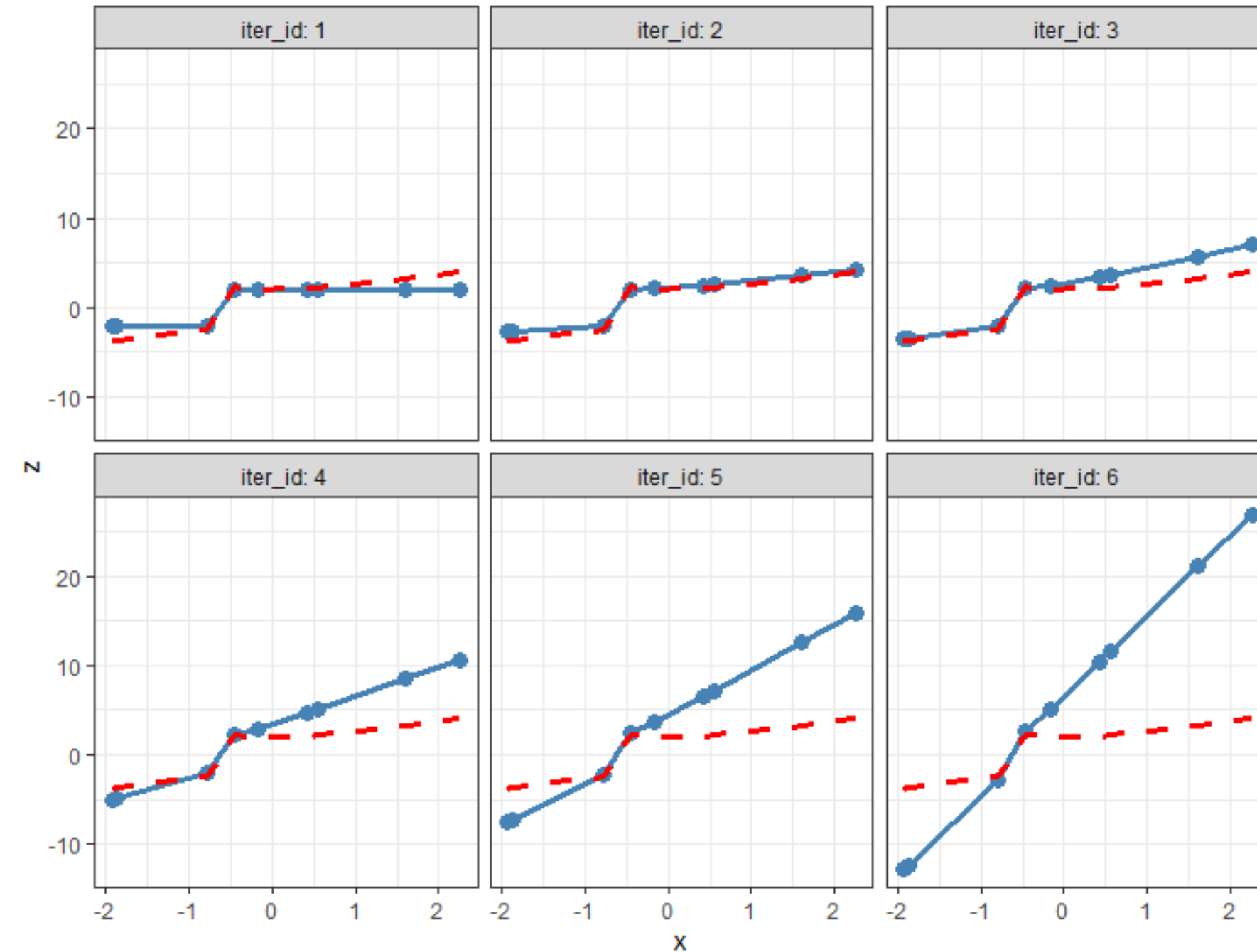




At the last iteration before the crash, event observations have μ 's very close to 1!

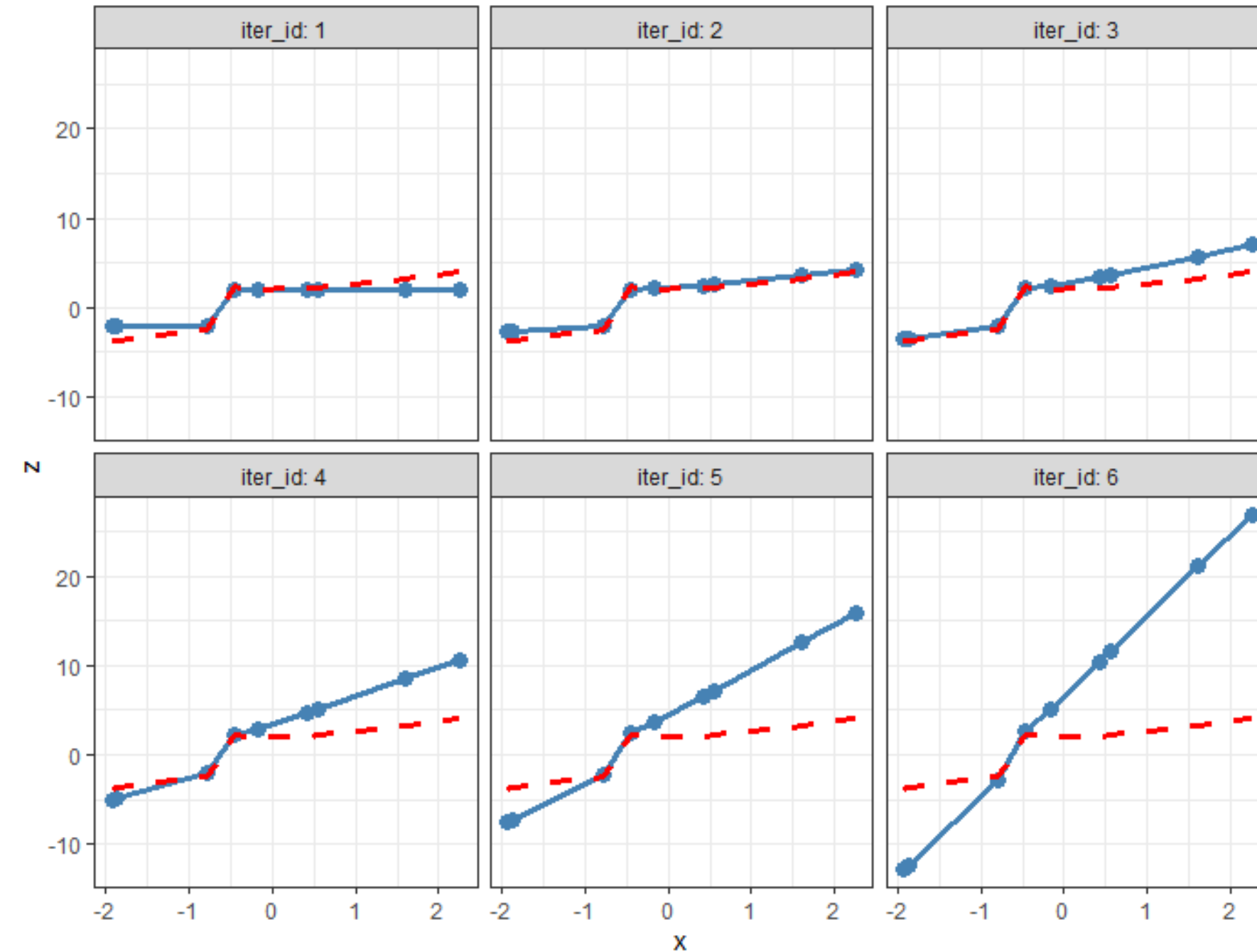
The μ 's are therefore nearly the same as y .





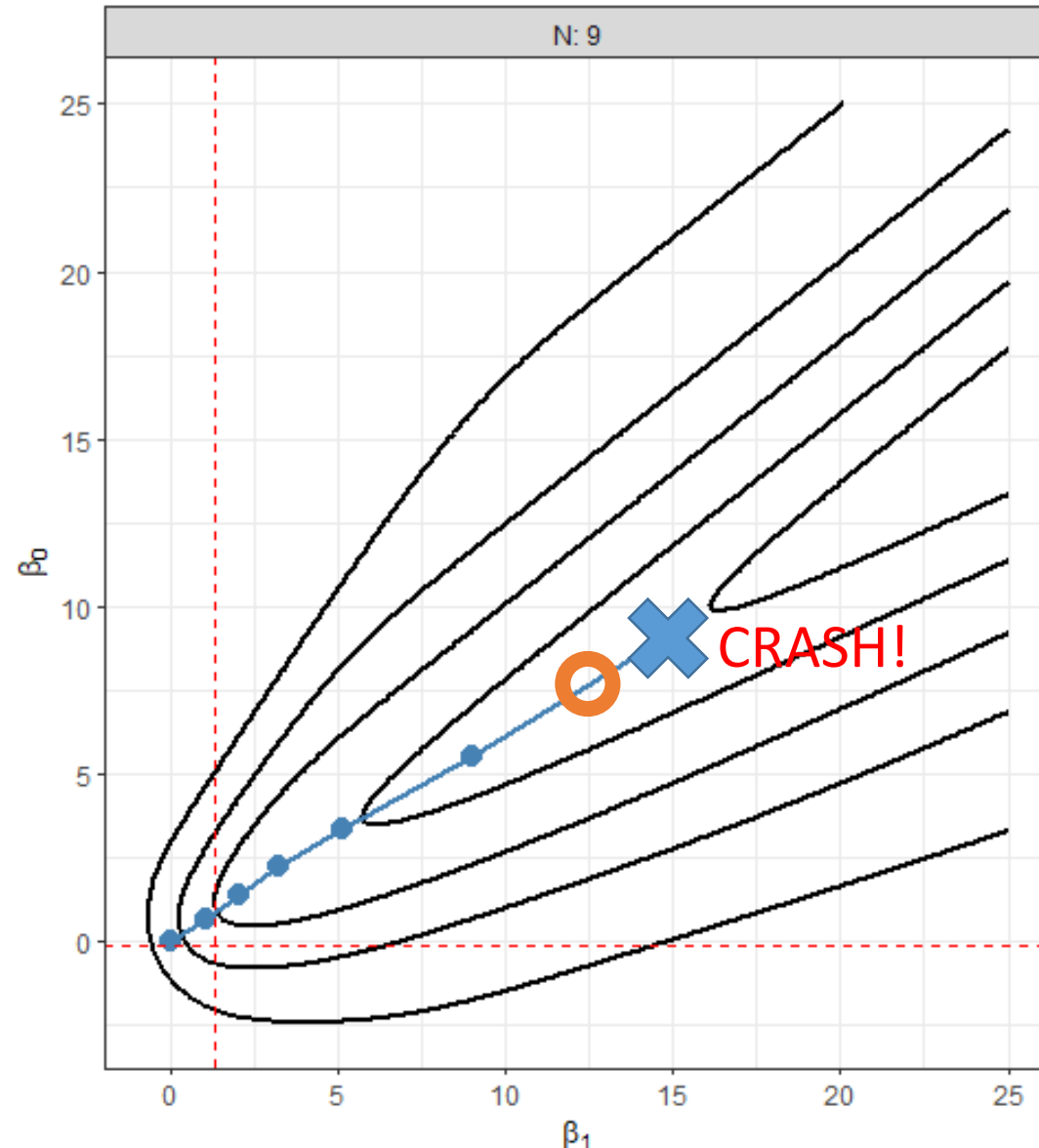
Since the μ 's are all consistent with the observed outcomes (the y 's) the adjustment factor contributes very little to the working response, z .

The linear predictor, η , completely controls z .



The working response is therefore unbounded!

Calculate the Hessian, and what the posterior covariance *would be* at a point just before the algorithm crashes



$$\text{Hessian} = -\mathbf{X}^T \mathbf{S} \mathbf{X}$$

	(Intercept)	x
(Intercept)	-0.2096555	0.12159028
x	0.1215903	-0.07654121

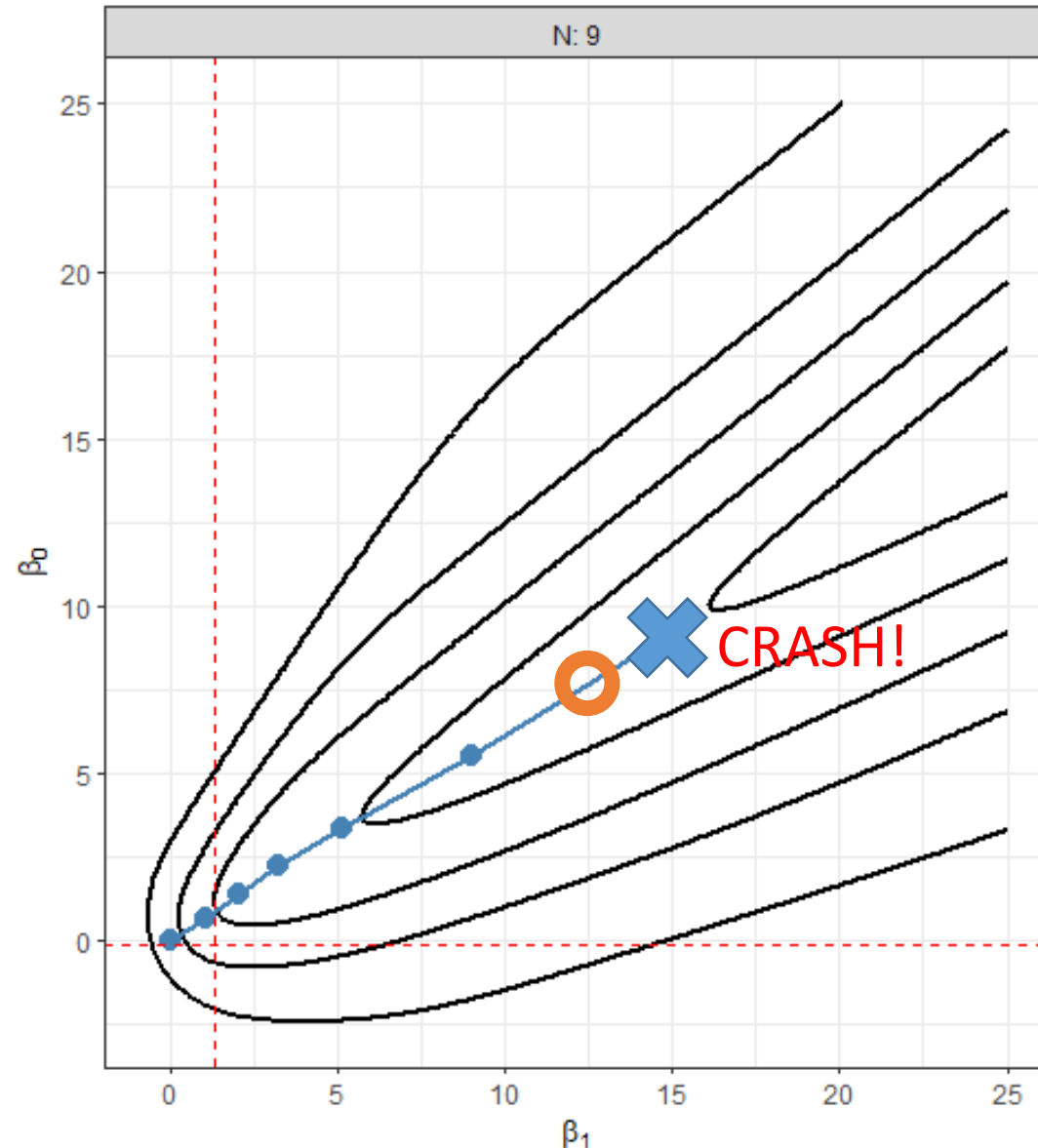
$$(\mathbf{X}^T \mathbf{S} \mathbf{X})^{-1}$$

	(Intercept)	x
(Intercept)	60.59848	96.26431
x	96.26431	165.98647

Cannot continue to keep “walking” in the direction of the mode.

The Hessian is not invertible beyond this point!

Calculate the Hessian, and what the posterior covariance would be at a point just before the algorithm crashes



$$\text{Hessian} = -\mathbf{X}^T \mathbf{S} \mathbf{X}$$

	(Intercept)	x
(Intercept)	-0.2096555	0.12159028
x	0.1215903	-0.07654121

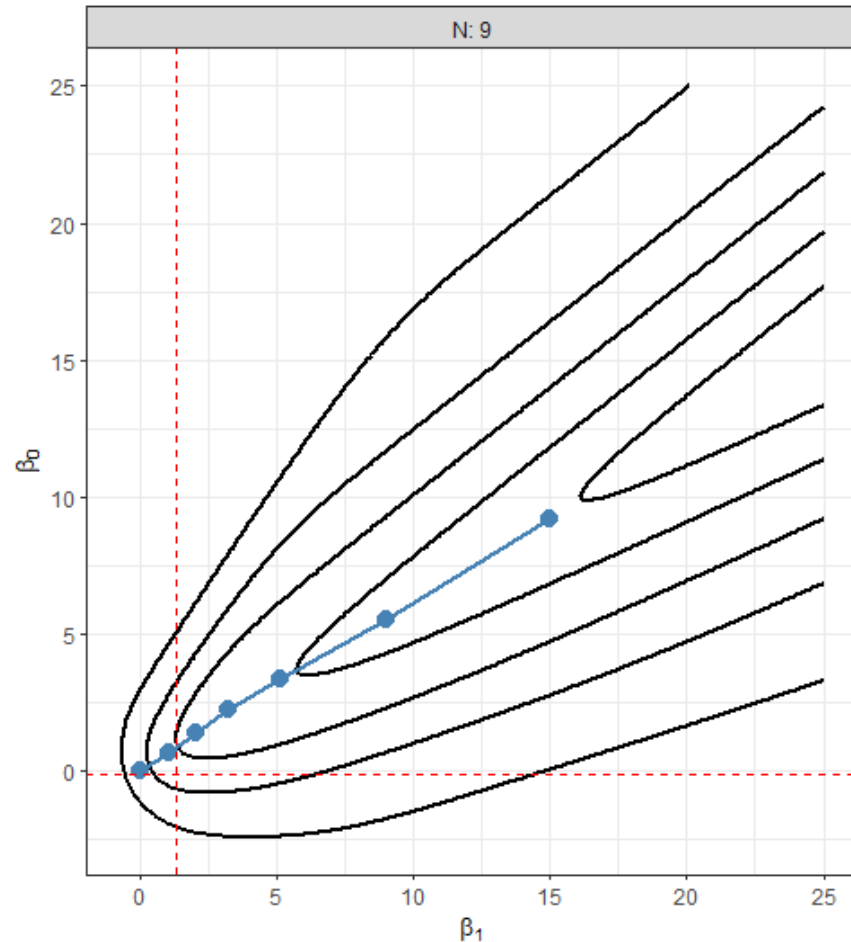
$$(\mathbf{X}^T \mathbf{S} \mathbf{X})^{-1}$$

	(Intercept)	x
(Intercept)	60.59848	96.26431
x	96.26431	165.98647

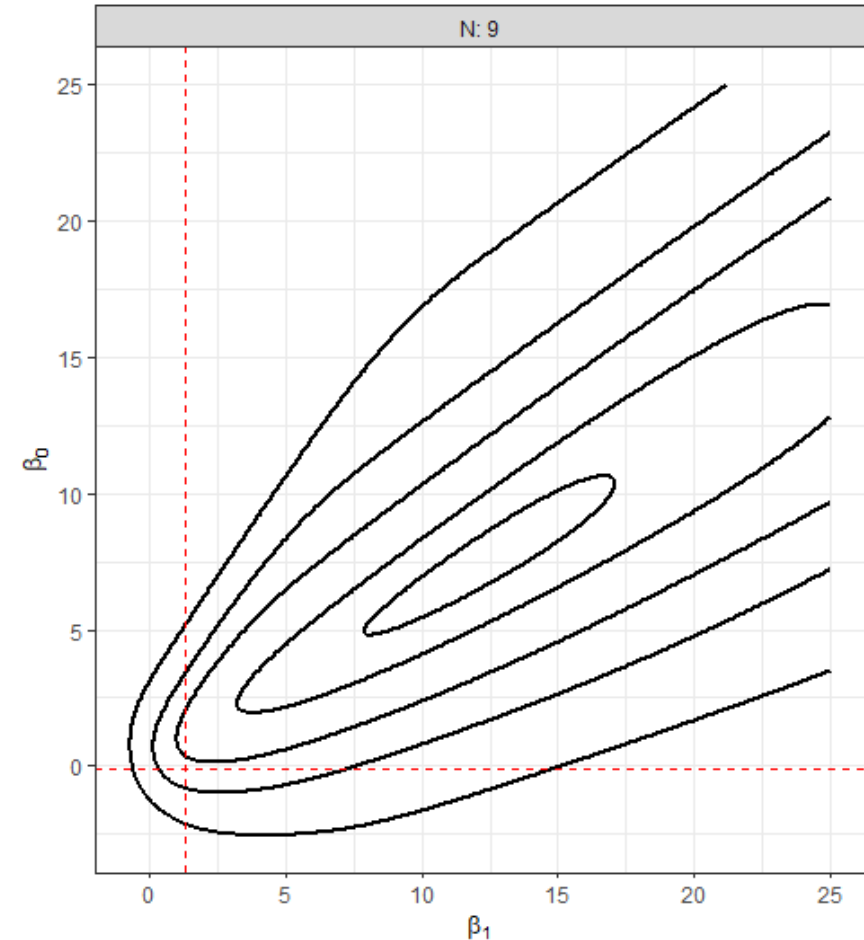
$$\text{Correlation}$$

	(Intercept)	x
(Intercept)	1.0000000	0.9598383
x	0.9598383	1.0000000

So what does the prior help us with?

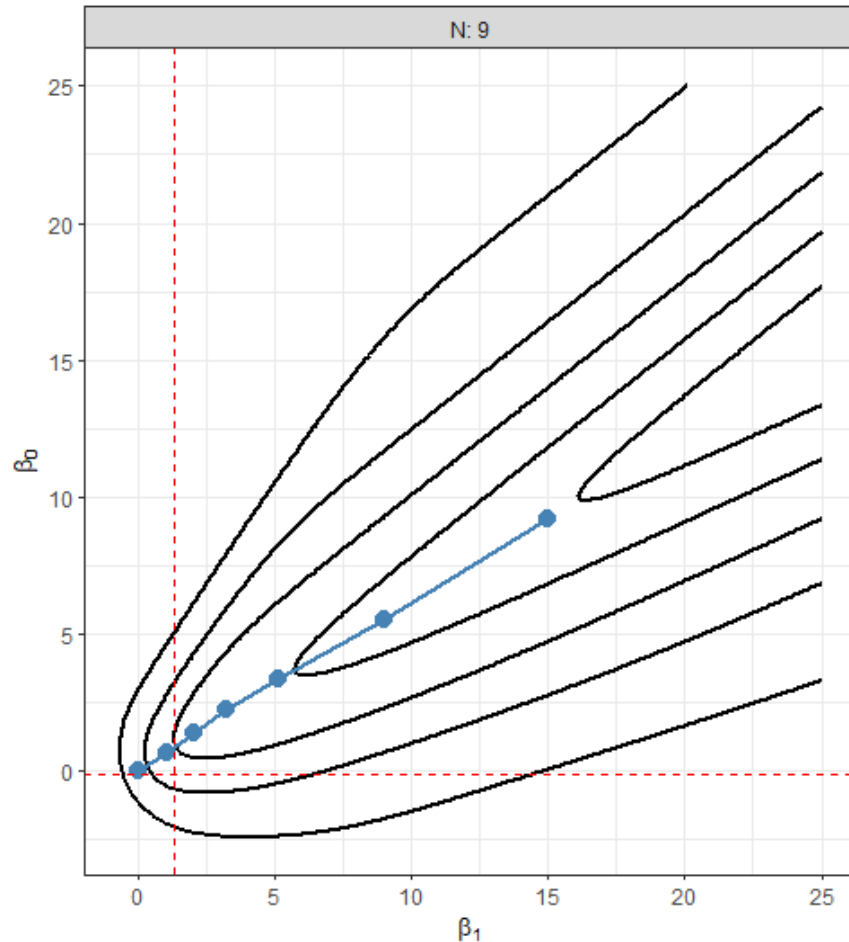


Log-likelihood

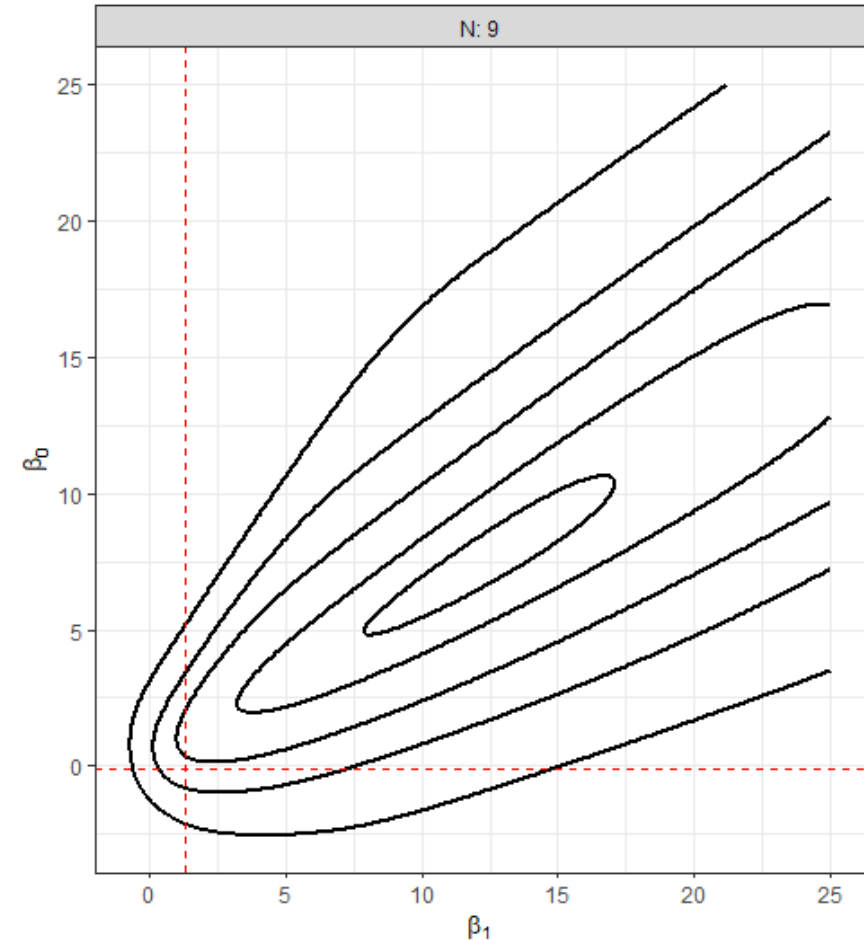


Log-posterior with a vague prior

Even a vague or “weak” prior “pulls” or *regularizes* the mode away from extreme coefficient values!



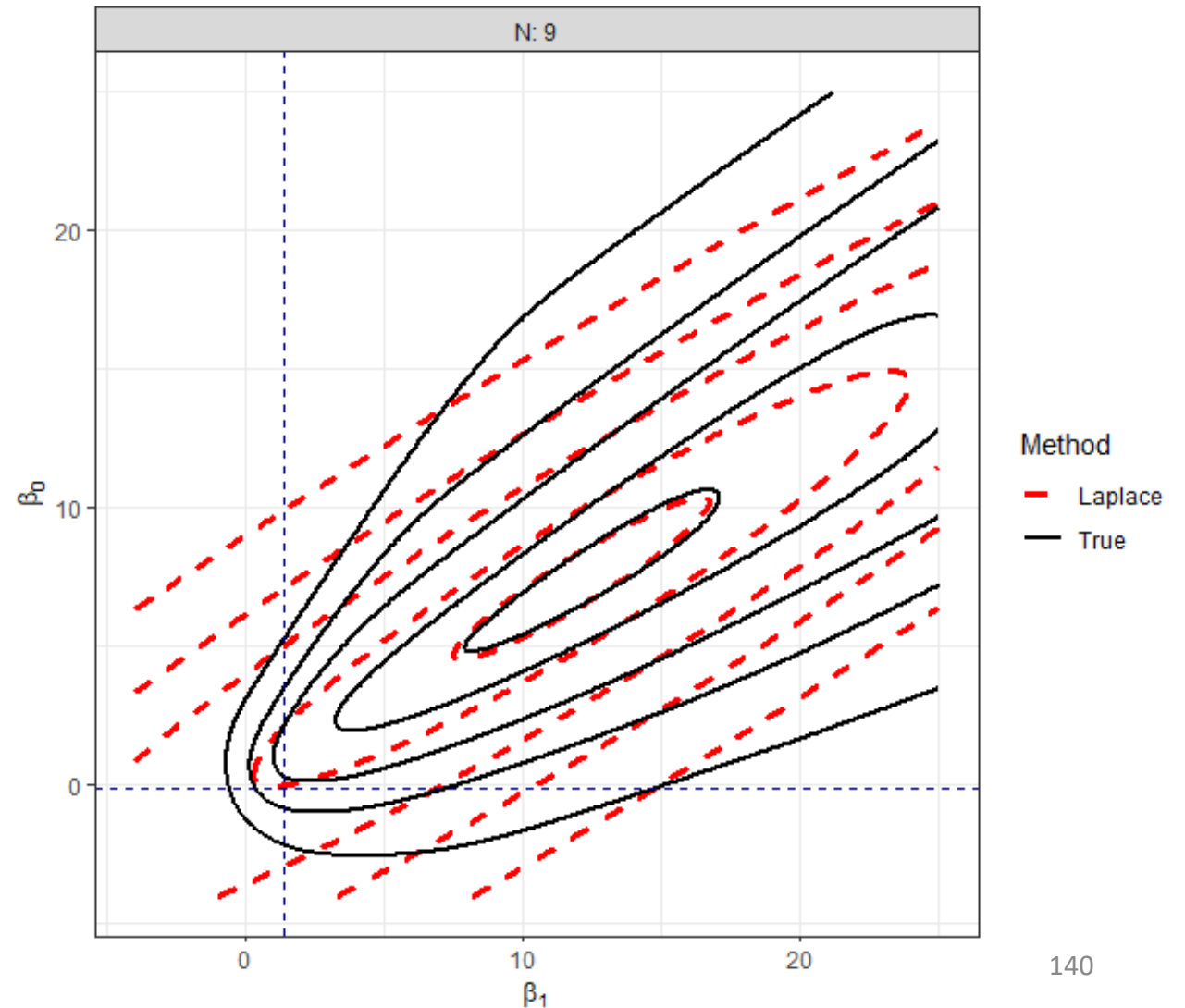
Log-likelihood



Log-posterior with a vague prior

Apply the Laplace Approximation to the case with 9 observations and the “weak” or diffuse prior

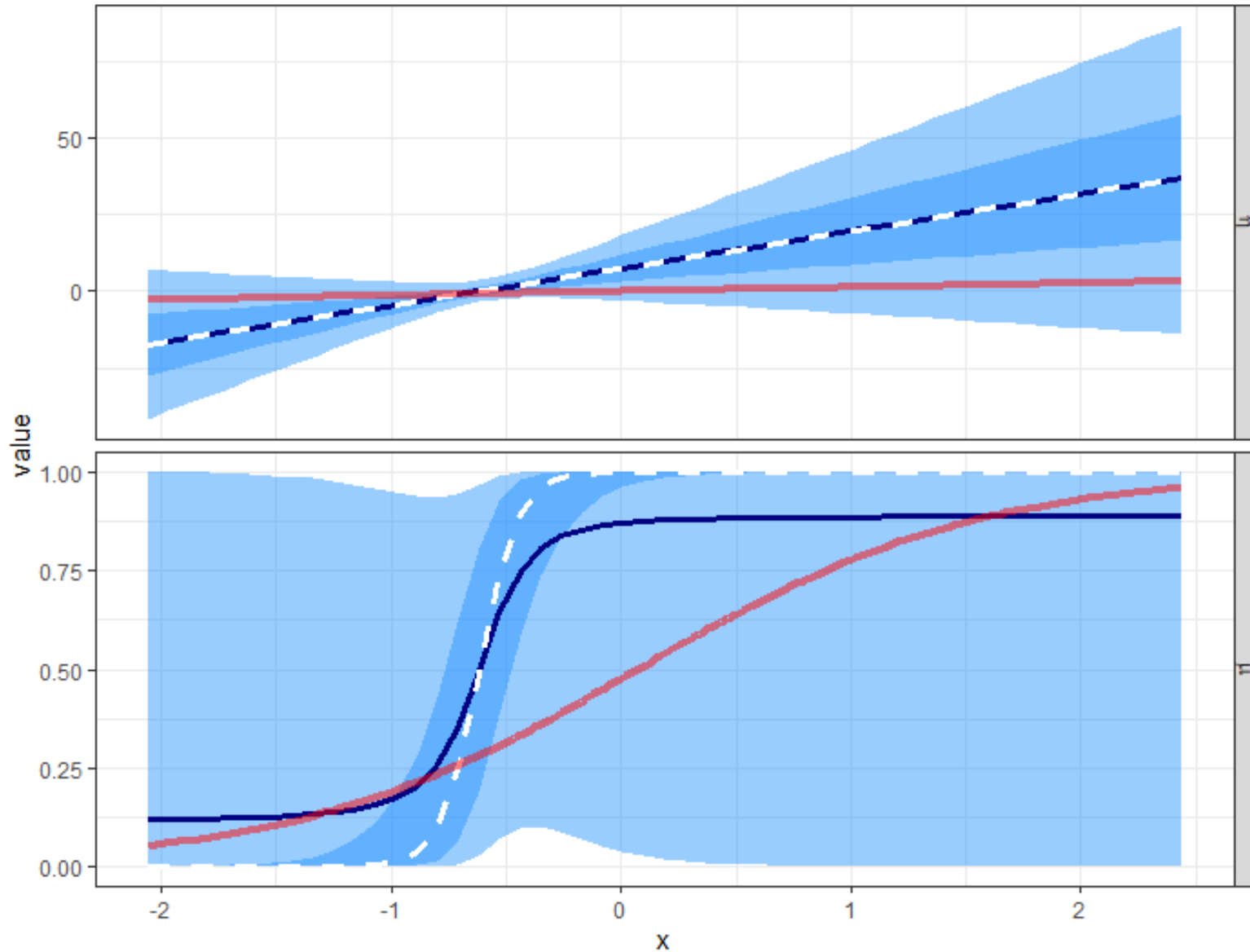
- Captures the mode quite well!
- Appears to over-estimate the uncertainty, based on the curvature around the mode compared to the true log-posterior.
- Less precise if the input, after 9 observations is “significant”.



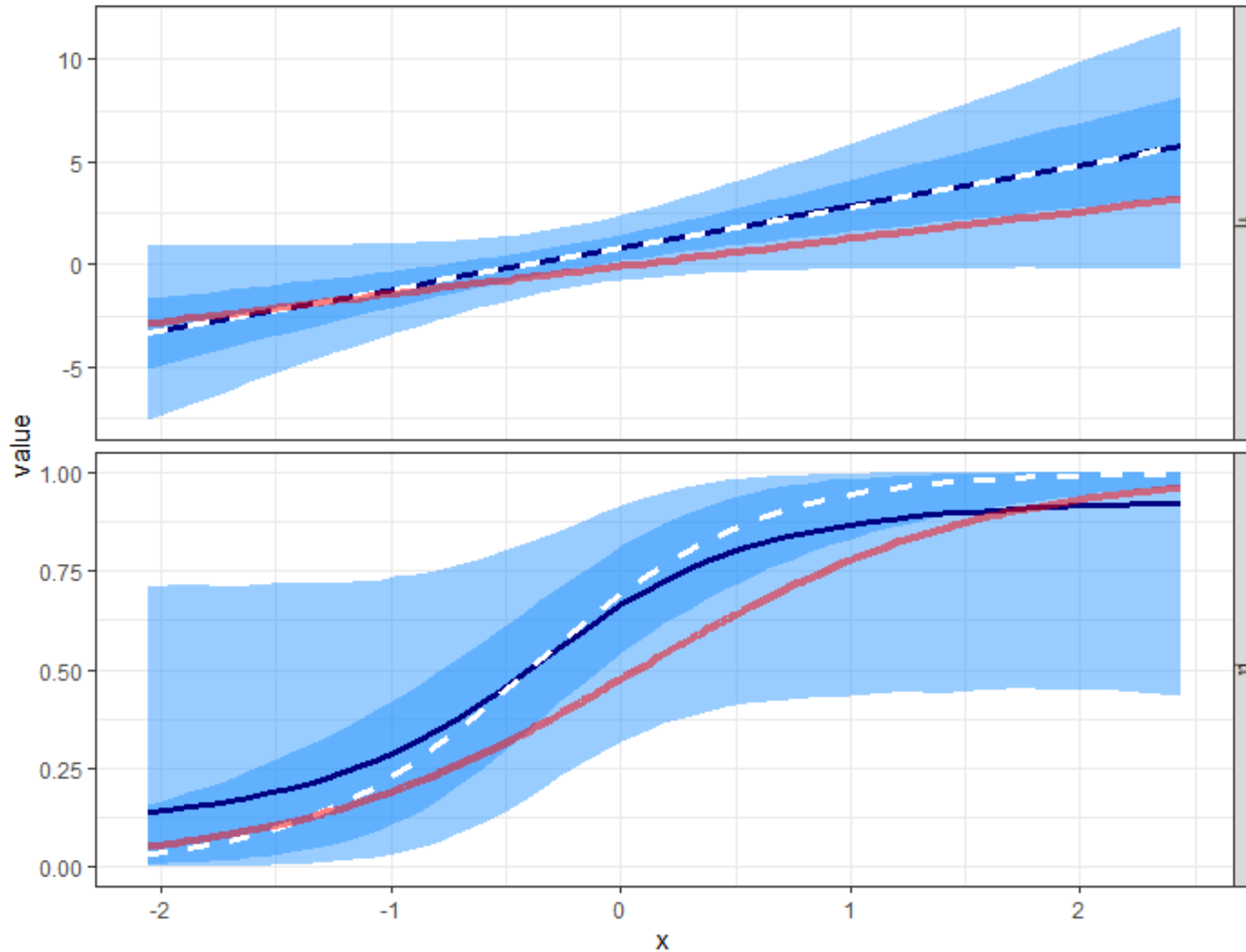
How do we make posterior predictions?

- Same setup as with linear models!
- Generate posterior samples of the β -parameters.
- Calculate the linear predictor for each posterior sample.
- Calculate the probability via the logistic function for each posterior sample.

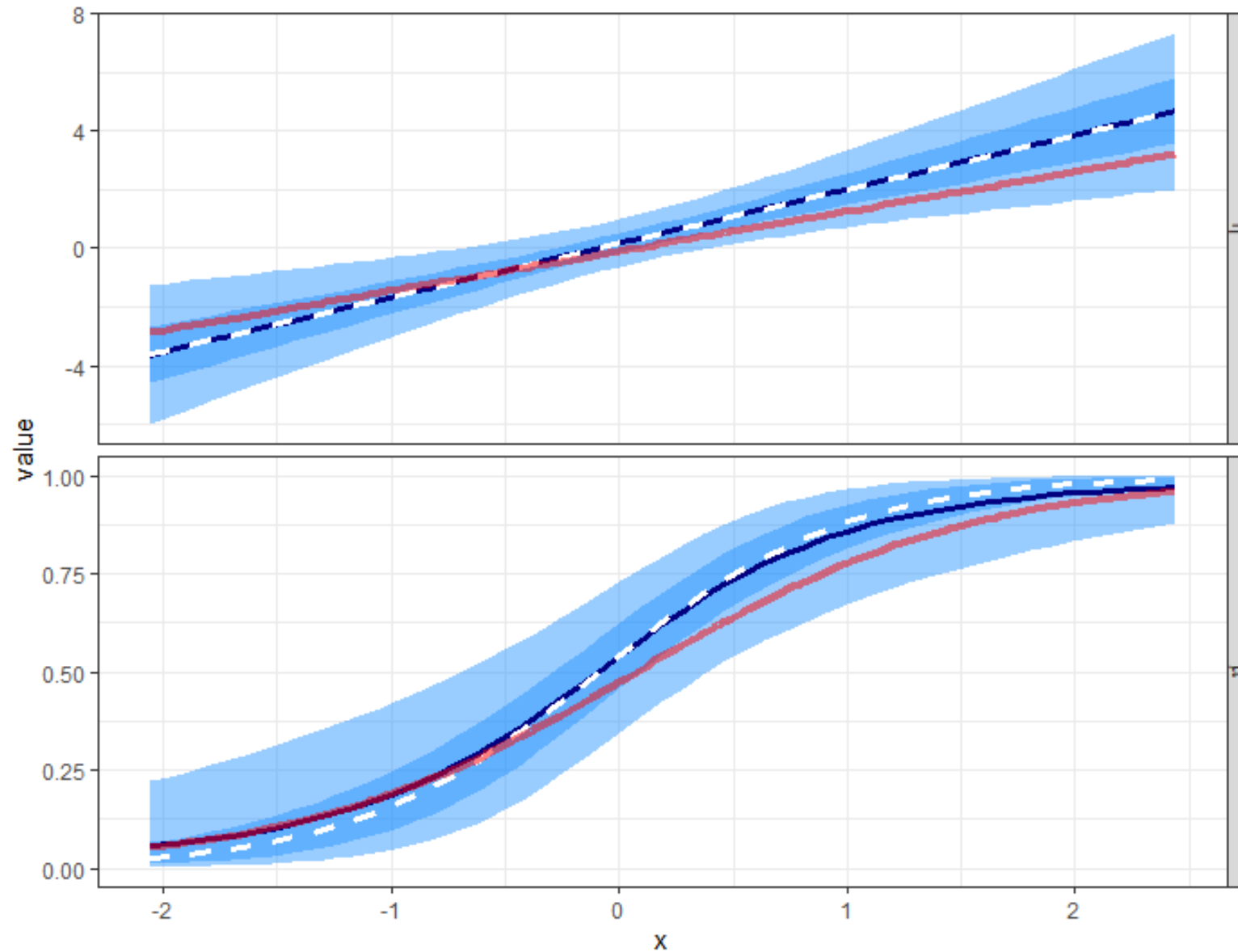
Posterior predictions based on 9 observations



Posterior predictions based on 10 observations



Posterior predictions based on 30 observations



Posterior predictions based on 50 observations

